

We can extend bit operations to bit strings. We define the **bitwise OR**, **bitwise AND**, and **bitwise XOR** of two strings of the same length to be the strings that have as their bits the *OR*, *AND*, and *XOR* of the corresponding bits in the two strings, respectively. We use the symbols $\vee$, $\wedge$, and $\oplus$ to represent the bitwise *OR*, bitwise *AND*, and bitwise *XOR* operations, respectively. We illustrate bitwise operations on bit strings with Example 16.

EXAMPLE 16 Find the bitwise *OR*, bitwise *AND*, and bitwise *XOR* of the bit strings 01 1011 0110 and 11 0001 1101. (Here, and throughout this book, bit strings will be split into blocks of four bits to make them easier to read.)

Solution: The bitwise *OR*, bitwise *AND*, and bitwise *XOR* of these strings are obtained by taking the *OR*, *AND*, and *XOR* of the corresponding bits, respectively. This gives us

01 1011 0110	
11 0001 1101	
11 1011 1111	bitwise <i>OR</i>
01 0001 0100	bitwise <i>AND</i>
10 1010 1011	bitwise <i>XOR</i>

Exercises

- Which of these sentences are propositions? What are the truth values of those that are propositions?
 - Boston is the capital of Massachusetts.
 - Miami is the capital of Florida.
 - $2 + 3 = 5$.
 - $5 + 7 = 10$.
 - $x + 2 = 11$.
 - Answer this question.
- Which of these are propositions? What are the truth values of those that are propositions?
 - Do not pass go.
 - What time is it?
 - There are no black flies in Maine.
 - $4 + x = 5$.
 - The moon is made of green cheese.
 - $2^n \geq 100$.
- What is the negation of each of these propositions?
 - Linda is younger than Sanjay.
 - Mei makes more money than Isabella.
 - Moshe is taller than Monica.
 - Abby is richer than Ricardo.
- What is the negation of each of these propositions?
 - Janice has more Facebook friends than Juan.
 - Quincy is smarter than Venkat.
 - Zelda drives more miles to school than Paola.
 - Briana sleeps longer than Gloria.
- What is the negation of each of these propositions?
 - Mei has an MP3 player.
 - There is no pollution in New Jersey.
 - $2 + 1 = 3$.
 - The summer in Maine is hot and sunny.
- What is the negation of each of these propositions?
 - Jennifer and Teja are friends.
 - There are 13 items in a baker's dozen.
 - Abby sent more than 100 text messages yesterday.
 - 121 is a perfect square.
- What is the negation of each of these propositions?
 - Steve has more than 100 GB free disk space on his laptop.
 - Zach blocks e-mails and texts from Jennifer.
 - $7 \cdot 11 \cdot 13 = 999$.
 - Diane rode her bicycle 100 miles on Sunday.
- Suppose that Smartphone A has 256 MB RAM and 32 GB ROM, and the resolution of its camera is 8 MP; Smartphone B has 288 MB RAM and 64 GB ROM, and the resolution of its camera is 4 MP; and Smartphone C has 128 MB RAM and 32 GB ROM, and the resolution of its camera is 5 MP. Determine the truth value of each of these propositions.
 - Smartphone B has the most RAM of these three smartphones.
 - Smartphone C has more ROM or a higher resolution camera than Smartphone B.
 - Smartphone B has more RAM, more ROM, and a higher resolution camera than Smartphone A.
 - If Smartphone B has more RAM and more ROM than Smartphone C, then it also has a higher resolution camera.
 - Smartphone A has more RAM than Smartphone B if and only if Smartphone B has more RAM than Smartphone A.
- Suppose that during the most recent fiscal year, the annual revenue of Acme Computer was 138 billion dollars and its net profit was 8 billion dollars, the annual revenue of Nadir Software was 87 billion dollars and its net profit was 5 billion dollars, and the annual revenue of Quixote Media was 111 billion dollars and its net profit was 13 billion dollars. Determine the truth value of each of these propositions for the most recent fiscal year.

- a) Quixote Media had the largest annual revenue.
- b) Nadir Software had the lowest net profit and Acme Computer had the largest annual revenue.
- c) Acme Computer had the largest net profit or Quixote Media had the largest net profit.
- d) If Quixote Media had the smallest net profit, then Acme Computer had the largest annual revenue.
- e) Nadir Software had the smallest net profit if and only if Acme Computer had the largest annual revenue.

10. Let p and q be the propositions p : I bought a lottery ticket this week. q : I won the million dollar jackpot.

Express each of these propositions as an English sentence.

- | | | |
|---------------------------|-------------------------------|--------------------------------|
| a) $\neg p$ | b) $p \vee q$ | c) $p \rightarrow q$ |
| d) $p \wedge q$ | e) $p \leftrightarrow q$ | f) $\neg p \rightarrow \neg q$ |
| g) $\neg p \wedge \neg q$ | h) $\neg p \vee (p \wedge q)$ | |

11. Let p and q be the propositions "Swimming at the New Jersey shore is allowed" and "Sharks have been spotted near the shore," respectively. Express each of these compound propositions as an English sentence.

- | | | |
|-------------------------------|------------------------------------|--------------------------------|
| a) $\neg q$ | b) $p \wedge q$ | c) $\neg p \vee q$ |
| d) $p \rightarrow \neg q$ | e) $\neg q \rightarrow p$ | f) $\neg p \rightarrow \neg q$ |
| g) $p \leftrightarrow \neg q$ | h) $\neg p \wedge (p \vee \neg q)$ | |

12. Let p and q be the propositions "The election is decided" and "The votes have been counted," respectively. Express each of these compound propositions as an English sentence.

- | | |
|--------------------------------|------------------------------------|
| a) $\neg p$ | b) $p \vee q$ |
| c) $\neg p \wedge q$ | d) $q \rightarrow p$ |
| e) $\neg q \rightarrow \neg p$ | f) $\neg p \rightarrow \neg q$ |
| g) $p \leftrightarrow q$ | h) $\neg q \vee (\neg p \wedge q)$ |

13. Let p and q be the propositions p : It is below freezing. q : It is snowing.

Write these propositions using p and q and logical connectives (including negations).

- a) It is below freezing and snowing.
- b) It is below freezing but not snowing.
- c) It is not below freezing and it is not snowing.
- d) It is either snowing or below freezing (or both).
- e) If it is below freezing, it is also snowing.
- f) Either it is below freezing or it is snowing, but it is not snowing if it is below freezing.
- g) That it is below freezing is necessary and sufficient for it to be snowing.

14. Let p , q , and r be the propositions p : You have the flu. q : You miss the final examination. r : You pass the course.

Express each of these propositions as an English sentence.

- | | |
|---|--|
| a) $p \rightarrow q$ | b) $\neg q \leftrightarrow r$ |
| c) $q \rightarrow \neg r$ | d) $p \vee q \vee r$ |
| e) $(p \rightarrow \neg r) \vee (q \rightarrow \neg r)$ | f) $(p \wedge q) \vee (\neg q \wedge r)$ |

15. Let p and q be the propositions p : You drive over 65 miles per hour. q : You get a speeding ticket.

Write these propositions using p and q and logical connectives (including negations).

- a) You do not drive over 65 miles per hour.
- b) You drive over 65 miles per hour, but you do not get a speeding ticket.
- c) You will get a speeding ticket if you drive over 65 miles per hour.
- d) If you do not drive over 65 miles per hour, then you will not get a speeding ticket.
- e) Driving over 65 miles per hour is sufficient for getting a speeding ticket.
- f) You get a speeding ticket, but you do not drive over 65 miles per hour.
- g) Whenever you get a speeding ticket, you are driving over 65 miles per hour.

16. Let p , q , and r be the propositions p : You get an A on the final exam. q : You do every exercise in this book. r : You get an A in this class.

Write these propositions using p , q , and r and logical connectives (including negations).

- a) You get an A in this class, but you do not do every exercise in this book.
- b) You get an A on the final, you do every exercise in this book, and you get an A in this class.
- c) To get an A in this class, it is necessary for you to get an A on the final.
- d) You get an A on the final, but you don't do every exercise in this book; nevertheless, you get an A in this class.
- e) Getting an A on the final and doing every exercise in this book is sufficient for getting an A in this class.
- f) You will get an A in this class if and only if you either do every exercise in this book or you get an A on the final.

17. Let p , q , and r be the propositions p : Grizzly bears have been seen in the area. q : Hiking is safe on the trail. r : Berries are ripe along the trail.

Write these propositions using p , q , and r and logical connectives (including negations).

- a) Berries are ripe along the trail, but grizzly bears have not been seen in the area.
- b) Grizzly bears have not been seen in the area and hiking on the trail is safe, but berries are ripe along the trail.
- c) If berries are ripe along the trail, hiking is safe if and only if grizzly bears have not been seen in the area.
- d) It is not safe to hike on the trail, but grizzly bears have not been seen in the area and the berries along the trail are ripe.
- e) For hiking on the trail to be safe, it is necessary but not sufficient that berries not be ripe along the trail and for grizzly bears not to have been seen in the area.

- f) Hiking is not safe on the trail whenever grizzly bears have been seen in the area and berries are ripe along the trail.
18. Determine whether these biconditionals are true or false.
- $2 + 2 = 4$ if and only if $1 + 1 = 2$.
 - $1 + 1 = 2$ if and only if $2 + 3 = 4$.
 - $1 + 1 = 3$ if and only if monkeys can fly.
 - $0 > 1$ if and only if $2 > 1$.
19. Determine whether each of these conditional statements is true or false.
- If $1 + 1 = 2$, then $2 + 2 = 5$.
 - If $1 + 1 = 3$, then $2 + 2 = 4$.
 - If $1 + 1 = 3$, then $2 + 2 = 5$.
 - If monkeys can fly, then $1 + 1 = 3$.
20. Determine whether each of these conditional statements is true or false.
- If $1 + 1 = 3$, then unicorns exist.
 - If $1 + 1 = 3$, then dogs can fly.
 - If $1 + 1 = 2$, then dogs can fly.
 - If $2 + 2 = 4$, then $1 + 2 = 3$.
21. For each of these sentences, determine whether an inclusive or, or an exclusive or, is intended. Explain your answer.
- Coffee or tea comes with dinner.
 - A password must have at least three digits or be at least eight characters long.
 - The prerequisite for the course is a course in number theory or a course in cryptography.
 - You can pay using U.S. dollars or euros.
22. For each of these sentences, determine whether an inclusive or, or an exclusive or, is intended. Explain your answer.
- Experience with C++ or Java is required.
 - Lunch includes soup or salad.
 - To enter the country you need a passport or a voter registration card.
 - Publish or perish.
23. For each of these sentences, state what the sentence means if the logical connective or is an inclusive or (that is, a disjunction) versus an exclusive or. Which of these meanings of or do you think is intended?
- To take discrete mathematics, you must have taken calculus or a course in computer science.
 - When you buy a new car from Acme Motor Company, you get \$2000 back in cash or a 2% car loan.
 - Dinner for two includes two items from column A or three items from column B.
 - School is closed if more than two feet of snow falls or if the wind chill is below -100°F .
24. Write each of these statements in the form “if p , then q ” in English. [*Hint*: Refer to the list of common ways to express conditional statements provided in this section.]
- It is necessary to wash the boss’s car to get promoted.
 - Winds from the south imply a spring thaw.
 - A sufficient condition for the warranty to be good is that you bought the computer less than a year ago.
 - Willy gets caught whenever he cheats.
 - You can access the website only if you pay a subscription fee.
 - Getting elected follows from knowing the right people.
 - Carol gets seasick whenever she is on a boat.
25. Write each of these statements in the form “if p , then q ” in English. [*Hint*: Refer to the list of common ways to express conditional statements.]
- It snows whenever the wind blows from the northeast.
 - The apple trees will bloom if it stays warm for a week.
 - That the Pistons win the championship implies that they beat the Lakers.
 - It is necessary to walk eight miles to get to the top of Long’s Peak.
 - To get tenure as a professor, it is sufficient to be world famous.
 - If you drive more than 400 miles, you will need to buy gasoline.
 - Your guarantee is good only if you bought your CD player less than 90 days ago.
 - Jan will go swimming unless the water is too cold.
 - We will have a future, provided that people believe in science.
26. Write each of these statements in the form “if p , then q ” in English. [*Hint*: Refer to the list of common ways to express conditional statements provided in this section.]
- I will remember to send you the address only if you send me an e-mail message.
 - To be a citizen of this country, it is sufficient that you were born in the United States.
 - If you keep your textbook, it will be a useful reference in your future courses.
 - The Red Wings will win the Stanley Cup if their goalie plays well.
 - That you get the job implies that you had the best credentials.
 - The beach erodes whenever there is a storm.
 - It is necessary to have a valid password to log on to the server.
 - You will reach the summit unless you begin your climb too late.
 - You will get a free ice cream cone, provided that you are among the first 100 customers tomorrow.
27. Write each of these propositions in the form “ p if and only if q ” in English.
- If it is hot outside you buy an ice cream cone, and if you buy an ice cream cone it is hot outside.
 - For you to win the contest it is necessary and sufficient that you have the only winning ticket.
 - You get promoted only if you have connections, and you have connections only if you get promoted.
 - If you watch television your mind will decay, and conversely.
 - The trains run late on exactly those days when I take it.

28. Write each of these propositions in the form “ p if and only if q ” in English.
- For you to get an A in this course, it is necessary and sufficient that you learn how to solve discrete mathematics problems.
 - If you read the newspaper every day, you will be informed, and conversely.
 - It rains if it is a weekend day, and it is a weekend day if it rains.
 - You can see the wizard only if the wizard is not in, and the wizard is not in only if you can see him.
 - My airplane flight is late exactly when I have to catch a connecting flight.
29. State the converse, contrapositive, and inverse of each of these conditional statements.
- If it snows today, I will ski tomorrow.
 - I come to class whenever there is going to be a quiz.
 - A positive integer is a prime only if it has no divisors other than 1 and itself.
30. State the converse, contrapositive, and inverse of each of these conditional statements.
- If it snows tonight, then I will stay at home.
 - I go to the beach whenever it is a sunny summer day.
 - When I stay up late, it is necessary that I sleep until noon.
31. How many rows appear in a truth table for each of these compound propositions?
- $p \rightarrow \neg p$
 - $(p \vee \neg r) \wedge (q \vee \neg s)$
 - $q \vee p \vee \neg s \vee \neg r \vee \neg t \vee u$
 - $(p \wedge r \wedge t) \leftrightarrow (q \wedge t)$
32. How many rows appear in a truth table for each of these compound propositions?
- $(q \rightarrow \neg p) \vee (\neg p \rightarrow \neg q)$
 - $(p \vee \neg t) \wedge (p \vee \neg s)$
 - $(p \rightarrow r) \vee (\neg s \rightarrow \neg t) \vee (\neg u \rightarrow v)$
 - $(p \wedge r \wedge s) \vee (q \wedge t) \vee (r \wedge \neg t)$
33. Construct a truth table for each of these compound propositions.
- $p \wedge \neg p$
 - $p \vee \neg p$
 - $(p \vee \neg q) \rightarrow q$
 - $(p \vee q) \rightarrow (p \wedge q)$
 - $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$
 - $(p \rightarrow q) \rightarrow (q \rightarrow p)$
34. Construct a truth table for each of these compound propositions.
- $p \rightarrow \neg p$
 - $p \leftrightarrow \neg p$
 - $p \oplus (p \vee q)$
 - $(p \wedge q) \rightarrow (p \vee q)$
 - $(q \rightarrow \neg p) \leftrightarrow (p \leftrightarrow q)$
 - $(p \leftrightarrow q) \oplus (p \leftrightarrow \neg q)$
35. Construct a truth table for each of these compound propositions.
- $(p \vee q) \rightarrow (p \oplus q)$
 - $(p \oplus q) \rightarrow (p \wedge q)$
 - $(p \vee q) \oplus (p \wedge q)$
 - $(p \leftrightarrow q) \oplus (\neg p \leftrightarrow q)$
 - $(p \leftrightarrow q) \oplus (\neg p \leftrightarrow \neg r)$
 - $(p \oplus q) \rightarrow (p \oplus \neg q)$
36. Construct a truth table for each of these compound propositions.
- $p \oplus p$
 - $p \oplus \neg p$
 - $p \oplus \neg q$
 - $\neg p \oplus \neg q$
 - $(p \oplus q) \vee (p \oplus \neg q)$
 - $(p \oplus q) \wedge (p \oplus \neg q)$
37. Construct a truth table for each of these compound propositions.
- $p \rightarrow \neg q$
 - $\neg p \leftrightarrow q$
 - $(p \rightarrow q) \vee (\neg p \rightarrow q)$
 - $(p \rightarrow q) \wedge (\neg p \rightarrow q)$
 - $(p \leftrightarrow q) \vee (\neg p \leftrightarrow q)$
 - $(\neg p \leftrightarrow \neg q) \leftrightarrow (p \leftrightarrow q)$
38. Construct a truth table for each of these compound propositions.
- $(p \vee q) \vee r$
 - $(p \vee q) \wedge r$
 - $(p \wedge q) \vee r$
 - $(p \wedge q) \wedge r$
 - $(p \vee q) \wedge \neg r$
 - $(p \wedge q) \vee \neg r$
39. Construct a truth table for each of these compound propositions.
- $p \rightarrow (\neg q \vee r)$
 - $\neg p \rightarrow (q \rightarrow r)$
 - $(p \rightarrow q) \vee (\neg p \rightarrow r)$
 - $(p \rightarrow q) \wedge (\neg p \rightarrow r)$
 - $(p \leftrightarrow q) \vee (\neg q \leftrightarrow r)$
 - $(\neg p \leftrightarrow \neg q) \leftrightarrow (q \leftrightarrow r)$
40. Construct a truth table for $((p \rightarrow q) \rightarrow r) \rightarrow s$.
41. Construct a truth table for $(p \leftrightarrow q) \leftrightarrow (r \leftrightarrow s)$.
42. Explain, without using a truth table, why $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$ is true when p, q , and r have the same truth value and it is false otherwise.
43. Explain, without using a truth table, why $(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ is true when at least one of p, q , and r is true and at least one is false, but is false when all three variables have the same truth value.
44. If $p_1, p_2, \dots, p_n$ are n propositions, explain why
- $$\bigwedge_{i=1}^{n-1} \bigwedge_{j=i+1}^n (\neg p_i \vee \neg p_j)$$
- is true if and only if at most one of $p_1, p_2, \dots, p_n$ is true.
45. Use Exercise 44 to construct a compound proposition that is true if and only if exactly one of the propositions $p_1, p_2, \dots, p_n$ is true. [Hint: Combine the compound proposition in Exercise 44 and a compound proposition that is true if and only if at least one of $p_1, p_2, \dots, p_n$ is true.]
46. What is the value of x after each of these statements is encountered in a computer program, if $x = 1$ before the statement is reached?
- if $x + 2 = 3$ then $x := x + 1$
 - if $(x + 1 = 3)$ OR $(2x + 2 = 3)$ then $x := x + 1$
 - if $(2x + 3 = 5)$ AND $(3x + 4 = 7)$ then $x := x + 1$
 - if $(x + 1 = 2)$ XOR $(x + 2 = 3)$ then $x := x + 1$
 - if $x < 2$ then $x := x + 1$
47. Find the bitwise OR, bitwise AND, and bitwise XOR of each of these pairs of bit strings.
- 101 1110, 010 0001
 - 1111 0000, 1010 1010
 - 00 0111 0001, 10 0100 1000
 - 11 1111 1111, 00 0000 0000

48. Evaluate each of these expressions.

- a) $1\ 1000 \wedge (0\ 1011 \vee 1\ 1011)$
- b) $(0\ 1111 \wedge 1\ 0101) \vee 0\ 1000$
- c) $(0\ 1010 \oplus 1\ 1011) \oplus 0\ 1000$
- d) $(1\ 1011 \vee 0\ 1010) \wedge (1\ 0001 \vee 1\ 1011)$

Fuzzy logic is used in artificial intelligence. In fuzzy logic, a proposition has a truth value that is a number between 0 and 1, inclusive. A proposition with a truth value of 0 is false and one with a truth value of 1 is true. Truth values that are between 0 and 1 indicate varying degrees of truth. For instance, the truth value 0.8 can be assigned to the statement “Fred is happy,” because Fred is happy most of the time, and the truth value 0.4 can be assigned to the statement “John is happy,” because John is happy slightly less than half the time. Use these truth values to solve Exercises 49–51.

- 49. The truth value of the negation of a proposition in fuzzy logic is 1 minus the truth value of the proposition. What are the truth values of the statements “Fred is not happy” and “John is not happy”?
- 50. The truth value of the conjunction of two propositions in fuzzy logic is the minimum of the truth values of the two propositions. What are the truth values of the statements

“Fred and John are happy” and “Neither Fred nor John is happy”?

- 51. The truth value of the disjunction of two propositions in fuzzy logic is the maximum of the truth values of the two propositions. What are the truth values of the statements “Fred is happy, or John is happy” and “Fred is not happy, or John is not happy”?
- *52. Is the assertion “This statement is false” a proposition?
- *53. The n th statement in a list of 100 statements is “Exactly n of the statements in this list are false.”
 - a) What conclusions can you draw from these statements?
 - b) Answer part (a) if the n th statement is “At least n of the statements in this list are false.”
 - c) Answer part (b) assuming that the list contains 99 statements.
- 54. An ancient Sicilian legend says that the barber in a remote town who can be reached only by traveling a dangerous mountain road shaves those people, and only those people, who do not shave themselves. Can there be such a barber?

1.2 Applications of Propositional Logic

1.2.1 Introduction

Logic has many important applications to mathematics, computer science, and numerous other disciplines. Statements in mathematics and the sciences and in natural language often are imprecise or ambiguous. To make such statements precise, they can be translated into the language of logic. For example, logic is used in the specification of software and hardware, because these specifications need to be precise before development begins. Furthermore, propositional logic and its rules can be used to design computer circuits, to construct computer programs, to verify the correctness of programs, and to build expert systems. Logic can be used to analyze and solve many familiar puzzles. Software systems based on the rules of logic have been developed for constructing some, but not all, types of proofs automatically. We will discuss some of these applications of propositional logic in this section and in later chapters.

1.2.2 Translating English Sentences

There are many reasons to translate English sentences into expressions involving propositional variables and logical connectives. In particular, English (and every other human language) is often ambiguous. Translating sentences into compound statements (and other types of logical expressions, which we will introduce later in this chapter) removes the ambiguity. Note that this may involve making a set of reasonable assumptions based on the intended meaning of the sentence. Moreover, once we have translated sentences from English into logical expressions, we can analyze these logical expressions to determine their truth values, we can manipulate them, and we can use rules of inference (which are discussed in Section 1.6) to reason about them.

To illustrate the process of translating an English sentence into a logical expression, consider Examples 1 and 2.

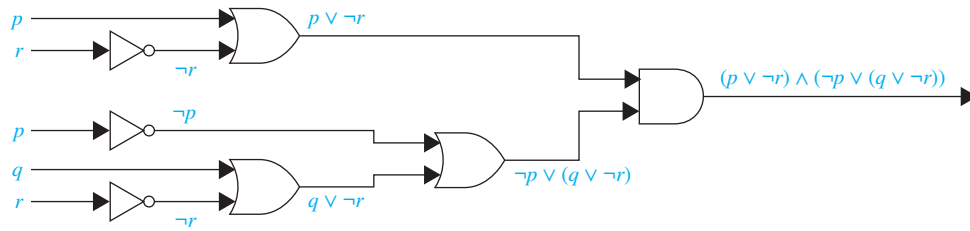


FIGURE 3 The circuit for $(p \vee \neg r) \wedge (\neg p \vee (q \vee \neg r))$.

EXAMPLE 11 Build a digital circuit that produces the output $(p \vee \neg r) \wedge (\neg p \vee (q \vee \neg r))$ when given input bits p , q , and r .

Solution: To construct the desired circuit, we build separate circuits for $p \vee \neg r$ and for $\neg p \vee (q \vee \neg r)$ and combine them using an AND gate. To construct a circuit for $p \vee \neg r$, we use an inverter to produce $\neg r$ from the input r . Then, we use an OR gate to combine p and $\neg r$. To build a circuit for $\neg p \vee (q \vee \neg r)$, we first use an inverter to obtain $\neg r$. Then we use an OR gate with inputs q and $\neg r$ to obtain $q \vee \neg r$. Finally, we use another inverter and an OR gate to get $\neg p \vee (q \vee \neg r)$ from the inputs p and $q \vee \neg r$.

To complete the construction, we employ a final AND gate, with inputs $p \vee \neg r$ and $\neg p \vee (q \vee \neg r)$. The resulting circuit is displayed in Figure 3. ◀

We will study logic circuits in great detail in Chapter 12 in the context of Boolean algebra, and with different notation.

Exercises

In Exercises 1–6, translate the given statement into propositional logic using the propositions provided.

1. You cannot edit a protected Wikipedia entry unless you are an administrator. Express your answer in terms of e : “You can edit a protected Wikipedia entry” and a : “You are an administrator.”
2. You can see the movie only if you are over 18 years old or you have the permission of a parent. Express your answer in terms of m : “You can see the movie,” e : “You are over 18 years old,” and p : “You have the permission of a parent.”
3. You can graduate only if you have completed the requirements of your major and you do not owe money to the university and you do not have an overdue library book. Express your answer in terms of g : “You can graduate,” m : “You owe money to the university,” r : “You have completed the requirements of your major,” and b : “You have an overdue library book.”
4. To use the wireless network in the airport you must pay the daily fee unless you are a subscriber to the service. Express your answer in terms of w : “You can use the wireless network in the airport,” d : “You pay the daily fee,” and s : “You are a subscriber to the service.”
5. You are eligible to be President of the U.S.A. only if you are at least 35 years old, were born in the U.S.A., or at the time of your birth both of your parents were citizens, and you have lived at least 14 years in the country. Express your answer in terms of e : “You are eligible to be President of the U.S.A.,” a : “You are at least 35 years old,” b : “You were born in the U.S.A.,” p : “At the time of your birth, both of your parents were citizens,” and r : “You have lived at least 14 years in the U.S.A.”
6. You can upgrade your operating system only if you have a 32-bit processor running at 1 GHz or faster, at least 1 GB RAM, and 16 GB free hard disk space, or a 64-bit processor running at 2 GHz or faster, at least 2 GB RAM, and at least 32 GB free hard disk space. Express your answer in terms of u : “You can upgrade your operating system,” b_{32} : “You have a 32-bit processor,” b_{64} : “You have a 64-bit processor,” g_1 : “Your processor runs at 1 GHz or faster,” g_2 : “Your processor runs at 2 GHz or faster,” r_1 : “Your processor has at least 1 GB RAM,” r_2 : “Your processor has at least 2 GB RAM,” h_{16} : “You have at least 16 GB free hard disk space,” and h_{32} : “You have at least 32 GB free hard disk space.”
7. Express these system specifications using the propositions p : “The message is scanned for viruses” and q : “The message was sent from an unknown system” together with logical connectives (including negations).
 - a) “The message is scanned for viruses whenever the message was sent from an unknown system.”

- b) "The message was sent from an unknown system but it was not scanned for viruses."
 - c) "It is necessary to scan the message for viruses whenever it was sent from an unknown system."
 - d) "When a message is not sent from an unknown system it is not scanned for viruses."
8. Express these system specifications using the propositions p : "The user enters a valid password," q : "Access is granted," and r : "The user has paid the subscription fee" and logical connectives (including negations).
- a) "The user has paid the subscription fee, but does not enter a valid password."
 - b) "Access is granted whenever the user has paid the subscription fee and enters a valid password."
 - c) "Access is denied if the user has not paid the subscription fee."
 - d) "If the user has not entered a valid password but has paid the subscription fee, then access is granted."
9. Are these system specifications consistent? "The system is in multiuser state if and only if it is operating normally. If the system is operating normally, the kernel is functioning. The kernel is not functioning or the system is in interrupt mode. If the system is not in multiuser state, then it is in interrupt mode. The system is not in interrupt mode."
10. Are these system specifications consistent? "Whenever the system software is being upgraded, users cannot access the file system. If users can access the file system, then they can save new files. If users cannot save new files, then the system software is not being upgraded."
11. Are these system specifications consistent? "The router can send packets to the edge system only if it supports the new address space. For the router to support the new address space it is necessary that the latest software release be installed. The router can send packets to the edge system if the latest software release is installed. The router does not support the new address space."
12. Are these system specifications consistent? "If the file system is not locked, then new messages will be queued. If the file system is not locked, then the system is functioning normally, and conversely. If new messages are not queued, then they will be sent to the message buffer. If the file system is not locked, then new messages will be sent to the message buffer. New messages will not be sent to the message buffer."
13. What Boolean search would you use to look for Web pages about beaches in New Jersey? What if you wanted to find Web pages about beaches on the isle of Jersey (in the English Channel)?
14. What Boolean search would you use to look for Web pages about hiking in West Virginia? What if you wanted to find Web pages about hiking in Virginia, but not in West Virginia?
15. What Google search would you use to look for Web pages relating to Ethiopian restaurants in New York or New Jersey?
16. What Google search would you use to look for men's shoes or boots not designed for work?
17. Suppose that in Example 7, the inscriptions on Trunks 1, 2, and 3 are "The treasure is in Trunk 3," "The treasure is in Trunk 1," and "This trunk is empty." For each of these statements, determine whether the Queen who never lies could state this, and if so, which trunk the treasure is in.
- a) "All the inscriptions are false."
 - b) "Exactly one of the inscriptions is true."
 - c) "Exactly two of the inscriptions are true."
 - d) "All three inscriptions are true."
18. Suppose that in Example 7 there are treasures in two of the three trunks. The inscriptions on Trunks 1, 2, and 3 are "This trunk is empty," "There is a treasure in Trunk 1," and "There is a treasure in Trunk 2." For each of these statements, determine whether the Queen who never lies could state this, and if so, which two trunks the treasures are in.
- a) "All the inscriptions are false."
 - b) "Exactly one of the inscriptions is true."
 - c) "Exactly two of the inscriptions are true."
 - d) "All three inscriptions are true."
- *19. Each inhabitant of a remote village always tells the truth or always lies. A villager will give only a "Yes" or a "No" response to a question a tourist asks. Suppose you are a tourist visiting this area and come to a fork in the road. One branch leads to the ruins you want to visit; the other branch leads deep into the jungle. A villager is standing at the fork in the road. What one question can you ask the villager to determine which branch to take?
20. An explorer is captured by a group of cannibals. There are two types of cannibals—those who always tell the truth and those who always lie. The cannibals will barbecue the explorer unless he can determine whether a particular cannibal always lies or always tells the truth. He is allowed to ask the cannibal exactly one question.
- a) Explain why the question "Are you a liar?" does not work.
 - b) Find a question that the explorer can use to determine whether the cannibal always lies or always tells the truth.
21. When three professors are seated in a restaurant, the hostess asks them: "Does everyone want coffee?" The first professor says "I do not know." The second professor then says "I do not know." Finally, the third professor says "No, not everyone wants coffee." The hostess comes back and gives coffee to the professors who want it. How did she figure out who wanted coffee?
22. When planning a party you want to know whom to invite. Among the people you would like to invite are three touchy friends. You know that if Jasmine attends, she will become unhappy if Samir is there, Samir will attend only if Kanti will be there, and Kanti will not attend unless Jasmine also does. Which combinations of these three friends can you invite so as not to make someone unhappy?

Exercises 23–27 relate to inhabitants of the island of knights and knaves created by Smullyan, where knights always tell the truth and knaves always lie. You encounter two people, *A* and *B*. Determine, if possible, what *A* and *B* are if they address you in the ways described. If you cannot determine what these two people are, can you draw any conclusions?

23. *A* says “At least one of us is a knave” and *B* says nothing.
24. *A* says “The two of us are both knights” and *B* says “*A* is a knave.”
25. *A* says “I am a knave or *B* is a knight” and *B* says nothing.
26. Both *A* and *B* say “I am a knight.”
27. *A* says “We are both knaves” and *B* says nothing.

Exercises 28–35 relate to inhabitants of an island on which there are three kinds of people: knights who always tell the truth, knaves who always lie, and spies (called normals by Smullyan [Sm78]) who can either lie or tell the truth. You encounter three people, *A*, *B*, and *C*. You know one of these people is a knight, one is a knave, and one is a spy. Each of the three people knows the type of person each of other two is. For each of these situations, if possible, determine whether there is a unique solution and determine who the knave, knight, and spy are. When there is no unique solution, list all possible solutions or state that there are no solutions.

28. *A* says “*C* is the knave,” *B* says “*A* is the knight,” and *C* says “I am the spy.”
29. *A* says “I am the knight,” *B* says “I am the knave,” and *C* says “*B* is the knight.”
30. *A* says “I am the knave,” *B* says “I am the knave,” and *C* says “I am the knave.”
31. *A* says “I am the knight,” *B* says “*A* is telling the truth,” and *C* says “I am the spy.”
32. *A* says “I am the knight,” *B* says “*A* is not the knave,” and *C* says “*B* is not the knave.”
33. *A* says “I am the knight,” *B* says “I am the knight,” and *C* says “I am the knight.”
34. *A* says “I am not the spy,” *B* says “I am not the spy,” and *C* says “*A* is the spy.”
35. *A* says “I am not the spy,” *B* says “I am not the spy,” and *C* says “I am not the spy.”

Exercises 36–42 are puzzles that can be solved by translating statements into logical expressions and reasoning from these expressions using truth tables.

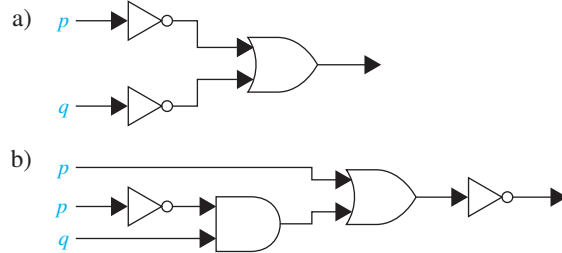
36. The police have three suspects for the murder of Mr. Cooper: Mr. Smith, Mr. Jones, and Mr. Williams. Smith, Jones, and Williams each declare that they did not kill Cooper. Smith also states that Cooper was a friend of Jones and that Williams disliked him. Jones also states that he did not know Cooper and that he was out of town the day Cooper was killed. Williams also states that he saw both Smith and Jones with Cooper the day of the killing and that either Smith or Jones must have killed him. Can you determine who the murderer was if
 - a) one of the three men is guilty, the two innocent men are telling the truth, but the statements of the guilty man may or may not be true?
 - b) innocent men do not lie?

37. Steve would like to determine the relative salaries of three coworkers using two facts. First, he knows that if Fred is not the highest paid of the three, then Janice is. Second, he knows that if Janice is not the lowest paid, then Maggie is paid the most. Is it possible to determine the relative salaries of Fred, Maggie, and Janice from what Steve knows? If so, who is paid the most and who the least? Explain your reasoning.
38. Five friends have access to a chat room. Is it possible to determine who is chatting if the following information is known? Either Kevin or Heather, or both, are chatting. Either Randy or Vijay, but not both, are chatting. If Abby is chatting, so is Randy. Vijay and Kevin are either both chatting or neither is. If Heather is chatting, then so are Abby and Kevin. Explain your reasoning.
39. A detective has interviewed four witnesses to a crime. From the stories of the witnesses the detective has concluded that if the butler is telling the truth then so is the cook; the cook and the gardener cannot both be telling the truth; the gardener and the handyman are not both lying; and if the handyman is telling the truth then the cook is lying. For each of the four witnesses, can the detective determine whether that person is telling the truth or lying? Explain your reasoning.
40. Four friends have been identified as suspects for an unauthorized access into a computer system. They have made statements to the investigating authorities. Alice said, “Carlos did it.” John said, “I did not do it.” Carlos said, “Diana did it.” Diana said, “Carlos lied when he said that I did it.”
 - a) If the authorities also know that exactly one of the four suspects is telling the truth, who did it? Explain your reasoning.
 - b) If the authorities also know that exactly one is lying, who did it? Explain your reasoning.
41. Suppose there are signs on the doors to two rooms. The sign on the first door reads “In this room there is a lady, and in the other one there is a tiger”; and the sign on the second door reads “In one of these rooms, there is a lady, and in one of them there is a tiger.” Suppose that you know that one of these signs is true and the other is false. Behind which door is the lady?
- *42. Solve this famous logic puzzle, attributed to Albert Einstein, and known as the **zebra puzzle**. Five men with different nationalities and with different jobs live in consecutive houses on a street. These houses are painted different colors. The men have different pets and have different favorite drinks. Determine who owns a zebra and whose favorite drink is mineral water (which is one of the favorite drinks) given these clues: The Englishman lives in the red house. The Spaniard owns a dog. The Japanese man is a painter. The Italian drinks tea. The Norwegian lives in the first house on the left. The green house is immediately to the right of the white one. The photographer breeds snails. The diplomat lives in the yellow house. Milk is drunk in the middle house. The owner of the green house drinks coffee. The Norwegian’s house is next to the

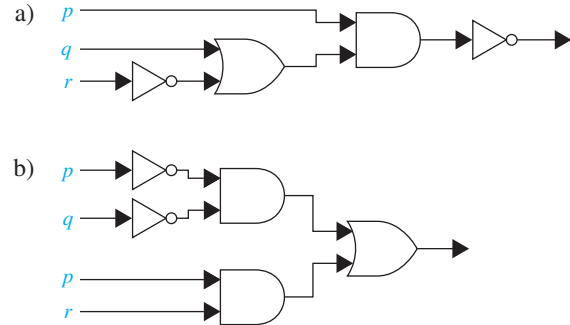


blue one. The violinist drinks orange juice. The fox is in a house next to that of the physician. The horse is in a house next to that of the diplomat. [Hint: Make a table where the rows represent the men and columns represent the color of their houses, their jobs, their pets, and their favorite drinks and use logical reasoning to determine the correct entries in the table.]

43. Freedonia has 50 senators. Each senator is either honest or corrupt. Suppose you know that at least one of the Freedonian senators is honest and that, given any two Freedonian senators, at least one is corrupt. Based on these facts, can you determine how many Freedonian senators are honest and how many are corrupt? If so, what is the answer?
44. Find the output of each of these combinatorial circuits.



45. Find the output of each of these combinatorial circuits.



46. Construct a combinatorial circuit using inverters, OR gates, and AND gates that produces the output $(p \wedge \neg r) \vee (\neg q \wedge r)$ from input bits p , q , and r .
47. Construct a combinatorial circuit using inverters, OR gates, and AND gates that produces the output $((\neg p \vee \neg r) \wedge \neg q) \vee (\neg p \wedge (q \vee r))$ from input bits p , q , and r .

1.3 Propositional Equivalences

1.3.1 Introduction

An important type of step used in a mathematical argument is the replacement of a statement with another statement with the same truth value. Because of this, methods that produce propositions with the same truth value as a given compound proposition are used extensively in the construction of mathematical arguments. Note that we will use the term “compound proposition” to refer to an expression formed from propositional variables using logical operators, such as $p \wedge q$.

We begin our discussion with a classification of compound propositions according to their possible truth values.

Definition 1

A compound proposition that is always true, no matter what the truth values of the propositional variables that occur in it, is called a *tautology*. A compound proposition that is always false is called a *contradiction*. A compound proposition that is neither a tautology nor a contradiction is called a *contingency*.

Tautologies and contradictions are often important in mathematical reasoning. Example 1 illustrates these types of compound propositions.

EXAMPLE 1 We can construct examples of tautologies and contradictions using just one propositional variable. Consider the truth tables of $p \vee \neg p$ and $p \wedge \neg p$, shown in Table 1. Because $p \vee \neg p$ is always true, it is a tautology. Because $p \wedge \neg p$ is always false, it is a contradiction. ◀

1.3.5 Satisfiability

A compound proposition is **satisfiable** if there is an assignment of truth values to its variables that makes it true (that is, when it is a tautology or a contingency). When no such assignments exists, that is, when the compound proposition is false for all assignments of truth values to its variables, the compound proposition is **unsatisfiable**. Note that a compound proposition is unsatisfiable if and only if its negation is true for all assignments of truth values to the variables, that is, if and only if its negation is a tautology.

When we find a particular assignment of truth values that makes a compound proposition true, we have shown that it is satisfiable; such an assignment is called a **solution** of this particular satisfiability problem. However, to show that a compound proposition is unsatisfiable, we need to show that *every* assignment of truth values to its variables makes it false. Although we can always use a truth table to determine whether a compound proposition is satisfiable, it is often more efficient not to, as Example 9 demonstrates.

EXAMPLE 9 Determine whether each of the compound propositions $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$, $(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$, and $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ is satisfiable.

Solution: Instead of using a truth table to solve this problem, we will reason about truth values. Note that $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$ is true when the three variables p , q , and r have the same truth value (see Exercise 42 of Section 1.1). Hence, it is satisfiable as there is at least one assignment of truth values for p , q , and r that makes it true. Similarly, note that $(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ is true when at least one of p , q , and r is true and at least one is false (see Exercise 43 of Section 1.1). Hence, $(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ is satisfiable, as there is at least one assignment of truth values for p , q , and r that makes it true.

Finally, note that for $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ to be true, $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$ and $(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ must both be true. For the first to be true, the three variables must have the same truth values, and for the second to be true, at least one of the three variables must be true and at least one must be false. However, these conditions are contradictory. From these observations we conclude that no assignment of truth values to p , q , and r makes $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ true. Hence, it is unsatisfiable. 

1.3.6 Applications of Satisfiability

Many problems, in diverse areas such as robotics, software testing, artificial intelligence planning, computer-aided design, machine vision, integrated circuit design, scheduling, computer networking, and genetics, can be modeled in terms of propositional satisfiability. Although most applications are quite complex and beyond the scope of this book, we can illustrate how two puzzles can be modeled as satisfiability problems.

EXAMPLE 10 The n -Queens Problem The n -queens problem asks for a placement of n queens on an $n \times n$ chessboard so that no queen can attack another queen. This means that no two queens can be placed in the same row, in the same column, or on the same diagonal. We display a solution to the eight-queens problem in Figure 1. (The eight-queens problem dates back to 1848 when it was proposed by Max Bezzel and was completely solved by Franz Nauck in 1850. We will return to the n -queens problem in Section 11.4.)

To model the n -queens problem as a satisfiability problem, we introduce n^2 variables, $p(i, j)$ for $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, n$. For a given placement of n queens on the chessboard, $p(i, j)$ is true when there is a queen on the square in the i th row and j th column, and is false

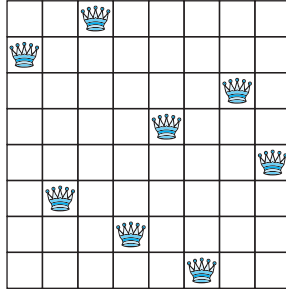


FIGURE 1

otherwise. Note that squares (i, j) and (i', j') are on the same diagonal if either $i + i' = j + j'$ or $i - i' = j - j'$. In the chessboard in Figure 1, $p(6, 2)$ and $p(2, 1)$ are true, while $p(3, 4)$ and $p(5, 4)$ are false.

For no two of the n queens to be in the same row, there must be one queen in each row. We can show that there is one queen in each row by verifying that every row contains at least one queen and that every row contains at most one queen. We first note that $\bigvee_{j=1}^n p(i, j)$ asserts that row i contains at least one queen, and

$$Q_1 = \bigwedge_{i=1}^n \bigvee_{j=1}^n p(i, j)$$

asserts that every row contains at least one queen.

For every row to include at most one queen, it must be the case that $p(i, j)$ and $p(k, j)$ are not both true for integers j and k with $1 \leq j < k \leq n$. Observe that $\neg p(i, j) \vee \neg p(k, j)$ asserts that at least one of $\neg p(i, j)$ and $\neg p(k, j)$ is true, which means that at least one of $p(i, j)$ and $p(k, j)$ is false. So, to check that there is at most one queen in each row, we assert

$$Q_2 = \bigwedge_{i=1}^n \bigwedge_{j=1}^{n-1} \bigwedge_{k=j+1}^n (\neg p(i, j) \vee \neg p(k, j)).$$

To assert that no column contains more than one queen, we assert that

$$Q_3 = \bigwedge_{j=1}^n \bigwedge_{i=1}^{n-1} \bigwedge_{k=i+1}^n (\neg p(i, j) \vee \neg p(k, j)).$$

(This assertion, together with the previous assertion that every row contains a queen, implies that every column contains a queen.)

To assert that no diagonal contains two queens, we assert

$$Q_4 = \bigwedge_{i=2}^n \bigwedge_{j=1}^{n-1} \bigwedge_{k=1}^{\min(i-1, n-j)} (\neg p(i, j) \vee \neg p(i-k, k+j))$$

and

$$Q_5 = \bigwedge_{i=1}^{n-1} \bigwedge_{j=1}^{n-1} \bigwedge_{k=1}^{\min(n-i, n-j)} (\neg p(i, j) \vee \neg p(i+k, j+k)).$$

The innermost conjunction in Q_4 and in Q_5 for a pair (i, j) runs through the positions on a diagonal that begin at (i, j) and runs rightward along this diagonal. The upper limits on these innermost conjunctions identify the last cell in the board on each diagonal.

Putting all this together, we find that the solutions of the n -queens problem are given by the assignments of truth values to the variables $p(i, j)$, $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, n$ that make

$$Q = Q_1 \wedge Q_2 \wedge Q_3 \wedge Q_4 \wedge Q_5$$

true.

Using this and other approaches, the number of ways n queens can be placed on a chessboard so that no queen can attack another has been computed for $n \leq 27$. When $n = 8$ there are 92 such placements, while for $n = 16$ this number grows to 14,772,512. (See the OEIS discussed in Section 2.4 for details.)

EXAMPLE 11 **Sudoku** Sudoku puzzles are constructed using a 9×9 grid made up of nine 3×3 subgrids, known as **blocks**, as shown in Figure 2. For each puzzle, some of the 81 cells, called **givens**, are assigned one of the numbers 1, 2, ..., 9, and the other cells are blank. The puzzle is solved by assigning a number to each blank cell so that every row, every column, and every one of the nine 3×3 blocks contains each of the nine possible numbers. Note that instead of using a 9×9 grid, Sudoku puzzles can be based on $n^2 \times n^2$ grids, for any positive integer n , with the $n^2 \times n^2$ grid made up of n^2 $n \times n$ subgrids.

Links

The popularity of Sudoku dates back to the 1980s when it was introduced in Japan. It took 20 years for Sudoku to spread to rest of the world, but by 2005, Sudoku puzzles were a worldwide craze. The name Sudoku is short for the Japanese *suuji wa dokushin ni kagiru*, which means “the digits must remain single.” The modern game of Sudoku was apparently designed in the late 1970s by an American puzzle designer. The basic ideas of Sudoku date back even further; puzzles printed in French newspapers in the 1890s were quite similar, but not identical, to modern Sudoku.

Sudoku puzzles designed for entertainment have two additional important properties. First, they have exactly one solution. Second, they can be solved using reasoning alone, that is, without resorting to searching all possible assignments of numbers to the cells. As a Sudoku puzzle is solved, entries in blank cells are successively determined by already known values. For instance, in the grid in Figure 2, the number 4 must appear in exactly one cell in the second row. How can we determine in which of the seven blank cells it must appear? First, we observe that 4 cannot appear in one of the first three cells or in one of the last three cells of this row, because it already appears in another cell in the block each of these cells is in. We can also see that 4

	2	9				4		
			5			1		
	4							
				4	2			
6							7	
5								
7			3					5
	1			9				
							6	

FIGURE 2 A 9×9 Sudoku puzzle.

cannot appear in the fifth cell in this row, as it already appears in the fifth column in the fourth row. This means that 4 must appear in the sixth cell of the second row.

Many strategies based on logic and mathematics have been devised for solving Sudoku puzzles (see [Da10], for example). Here, we discuss one of the ways that have been developed for solving Sudoku puzzles with the aid of a computer, which depends on modeling the puzzle as a propositional satisfiability problem. Using the model we describe, particular Sudoku puzzles can be solved using software developed to solve satisfiability problems. Currently, Sudoku puzzles can be solved in less than 10 milliseconds this way. It should be noted that there are many other approaches for solving Sudoku puzzles via computers using other techniques.

To encode a Sudoku puzzle, let $p(i, j, n)$ denote the proposition that is true when the number n is in the cell in the i th row and j th column. There are $9 \times 9 \times 9 = 729$ such propositions, as i , j , and n all range from 1 to 9. For example, for the puzzle in Figure 2, the number 6 is given as the value in the fifth row and first column. Hence, we see that $p(5, 1, 6)$ is true, but $p(5, j, 6)$ is false for $j = 2, 3, \dots, 9$.

Given a particular Sudoku puzzle, we begin by encoding each of the given values. Then, we construct compound propositions that assert that every row contains every number, every column contains every number, every 3×3 block contains every number, and each cell contains no more than one number. It follows, as the reader should verify, that the Sudoku puzzle is solved by finding an assignment of truth values to the 729 propositions $p(i, j, n)$ with i , j , and n each ranging from 1 to 9 that makes the conjunction of all these compound propositions true. After listing these assertions, we will explain how to construct the assertion that every row contains every integer from 1 to 9. We will leave the construction of the other assertions that every column contains every number and each of the nine 3×3 blocks contains every number to the exercises.

- ▶ For each cell with a given value, we assert $p(i, j, n)$ when the cell in row i and column j has the given value n .
- ▶ We assert that every row contains every number:

$$\bigwedge_{i=1}^9 \bigwedge_{n=1}^9 \bigvee_{j=1}^9 p(i, j, n)$$

- ▶ We assert that every column contains every number:

$$\bigwedge_{j=1}^9 \bigwedge_{n=1}^9 \bigvee_{i=1}^9 p(i, j, n)$$

- ▶ We assert that each of the nine 3×3 blocks contains every number:

$$\bigwedge_{r=0}^2 \bigwedge_{s=0}^2 \bigwedge_{n=1}^9 \bigvee_{i=1}^3 \bigvee_{j=1}^3 p(3r + i, 3s + j, n)$$

- ▶ To assert that no cell contains more than one number, we take the conjunction over all values of n, n', i , and j , where each variable ranges from 1 to 9 and $n \neq n'$ of $p(i, j, n) \rightarrow \neg p(i, j, n')$.

We now explain how to construct the assertion that every row contains every number. First, to assert that row i contains the number n , we form $\bigvee_{j=1}^9 p(i, j, n)$. To assert that row i contains all n numbers, we form the conjunction of these disjunctions over all nine possible values of n , giving us $\bigwedge_{n=1}^9 \bigvee_{j=1}^9 p(i, j, n)$. Finally, to assert that every row contains



It is tricky setting up the two inner indices so that all nine cells in each square block are examined.

every number, we take the conjunction of $\bigwedge_{n=1}^9 \bigvee_{j=1}^9 p(i, j, n)$ over all nine rows. This gives us $\bigwedge_{i=1}^9 \bigwedge_{n=1}^9 \bigvee_{j=1}^9 p(i, j, n)$. (Exercises 71 and 72 ask for explanations of the assertions that every column contains every number and that each of the nine 3×3 blocks contains every number.)

Given a particular Sudoku puzzle, to solve this puzzle we can find a solution to the satisfiability problems that asks for a set of truth values for the 729 variables $p(i, j, n)$ that makes the conjunction of all the listed assertions true. ▶

1.3.7 Solving Satisfiability Problems

A truth table can be used to determine whether a compound proposition is satisfiable, or equivalently, whether its negation is a tautology (see Exercise 64). This can be done by hand for a compound proposition with a small number of variables, but when the number of variables grows, this becomes impractical. For instance, there are $2^{20} = 1,048,576$ rows in the truth table for a compound proposition with 20 variables. Thus, you need a computer to help you determine, in this way, whether a compound proposition in 20 variables is satisfiable.

When many applications are modeled, questions concerning the satisfiability of compound propositions with hundreds, thousands, or millions of variables arise. Note, for example, that when there are 1000 variables, checking every one of the 2^{1000} (a number with more than 300 decimal digits) possible combinations of truth values of the variables in a compound proposition cannot be done by a computer in even trillions of years. No procedure is known that a computer can follow to determine in a reasonable amount of time whether an arbitrary compound proposition in such a large number of variables is satisfiable. However, progress has been made developing methods for solving the satisfiability problem for the particular types of compound propositions that arise in practical applications, such as for the solution of Sudoku puzzles. Many computer programs have been developed for solving satisfiability problems which have practical use. In our discussion of the subject of algorithms in Chapter 3, we will discuss this question further. In particular, we will explain the important role the propositional satisfiability problem plays in the study of the complexity of algorithms.

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Courtesy of Harvard University Portrait Collection, Department of Philosophy

HENRY MAURICE SHEFFER (1883–1964) Henry Maurice Sheffer, born to Jewish parents in the western Ukraine, emigrated to the United States in 1892 with his parents and six siblings. He studied at the Boston Latin School before entering Harvard, where he completed his undergraduate degree in 1905, his master's in 1907, and his Ph.D. in philosophy in 1908. After holding a postdoctoral position at Harvard, Henry traveled to Europe on a fellowship. Upon returning to the United States, he became an academic nomad, spending one year each at the University of Washington, Cornell, the University of Minnesota, the University of Missouri, and City College in New York. In 1916 he returned to Harvard as a faculty member in the philosophy department. He remained at Harvard until his retirement in 1952.

Sheffer introduced what is now known as the Sheffer stroke in 1913; it became well known only after its use in the 1925 edition of Whitehead and Russell's *Principia Mathematica*. In this same edition Russell wrote that Sheffer had invented a powerful method that could be used to simplify the *Principia*. Because of this comment, Sheffer was something of a mystery man to logicians, especially because Sheffer, who published

little in his career, never published the details of this method, only describing it in mimeographed notes and in a brief published abstract.

Sheffer was a dedicated teacher of mathematical logic. He liked his classes to be small and did not like auditors. When strangers appeared in his classroom, Sheffer would order them to leave, even his colleagues or distinguished guests visiting Harvard. Sheffer was barely five feet tall; he was noted for his wit and vigor, as well as for his nervousness and irritability. Although widely liked, he was quite lonely. He is noted for a quip he spoke at his retirement: "Old professors never die, they just become emeriti." Sheffer is also credited with coining the term "Boolean algebra" (the subject of Chapter 12 of this text). Sheffer was briefly married and lived most of his later life in small rooms at a hotel packed with his logic books and vast files of slips of paper he used to jot down his ideas. Unfortunately, Sheffer suffered from severe depression during the last two decades of his life.

Exercises

1. Use truth tables to verify these equivalences.
 - a) $p \wedge \mathbf{T} \equiv p$
 - b) $p \vee \mathbf{F} \equiv p$
 - c) $p \wedge \mathbf{F} \equiv \mathbf{F}$
 - d) $p \vee \mathbf{T} \equiv \mathbf{T}$
 - e) $p \vee p \equiv p$
 - f) $p \wedge p \equiv p$
2. Show that $\neg(\neg p)$ and p are logically equivalent.
3. Use truth tables to verify the commutative laws
 - a) $p \vee q \equiv q \vee p$.
 - b) $p \wedge q \equiv q \wedge p$.
4. Use truth tables to verify the associative laws
 - a) $(p \vee q) \vee r \equiv p \vee (q \vee r)$.
 - b) $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$.
5. Use a truth table to verify the distributive law
 $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$.
6. Use a truth table to verify the first De Morgan law
 $\neg(p \wedge q) \equiv \neg p \vee \neg q$.
7. Use De Morgan's laws to find the negation of each of the following statements.
 - a) Jan is rich and happy.
 - b) Carlos will bicycle or run tomorrow.
 - c) Mei walks or takes the bus to class.
 - d) Ibrahim is smart and hard working.
8. Use De Morgan's laws to find the negation of each of the following statements.
 - a) Kwame will take a job in industry or go to graduate school.
 - b) Yoshiko knows Java and calculus.
 - c) James is young and strong.
 - d) Rita will move to Oregon or Washington.
9. For each of these compound propositions, use the conditional-disjunction equivalence (Example 3) to find an equivalent compound proposition that does not involve conditionals.
 - a) $p \rightarrow \neg q$
 - b) $(p \rightarrow q) \rightarrow r$
 - c) $(\neg q \rightarrow p) \rightarrow (p \rightarrow \neg q)$
10. For each of these compound propositions, use the conditional-disjunction equivalence (Example 3) to find an equivalent compound proposition that does not involve conditionals.
 - a) $\neg p \rightarrow \neg q$
 - b) $(p \vee q) \rightarrow \neg p$
 - c) $(p \rightarrow \neg q) \rightarrow (\neg p \rightarrow q)$
-  11. Show that each of these conditional statements is a tautology by using truth tables.
 - a) $(p \wedge q) \rightarrow p$
 - b) $p \rightarrow (p \vee q)$
 - c) $\neg p \rightarrow (p \rightarrow q)$
 - d) $(p \wedge q) \rightarrow (p \rightarrow q)$
 - e) $\neg(p \rightarrow q) \rightarrow p$
 - f) $\neg(p \rightarrow q) \rightarrow \neg q$
-  12. Show that each of these conditional statements is a tautology by using truth tables.
 - a) $[\neg p \wedge (p \vee q)] \rightarrow q$
 - b) $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$
 - c) $[p \wedge (p \rightarrow q)] \rightarrow q$
 - d) $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$
13. Show that each conditional statement in Exercise 11 is a tautology using the fact that a conditional statement is false exactly when the hypothesis is true and the conclusion is false. (Do not use truth tables.)
14. Show that each conditional statement in Exercise 12 is a tautology using the fact that a conditional statement is false exactly when the hypothesis is true and the conclusion is false. (Do not use truth tables.)
15. Show that each conditional statement in Exercise 11 is a tautology by applying a chain of logical identities as in Example 8. (Do not use truth tables.)
16. Show that each conditional statement in Exercise 12 is a tautology by applying a chain of logical identities as in Example 8. (Do not use truth tables.)
17. Use truth tables to verify the absorption laws.
 - a) $p \vee (p \wedge q) \equiv p$
 - b) $p \wedge (p \vee q) \equiv p$
18. Determine whether $(\neg p \wedge (p \rightarrow q)) \rightarrow \neg q$ is a tautology.
-  19. Determine whether $(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$ is a tautology.
 Each of Exercises 20–32 asks you to show that two compound propositions are logically equivalent. To do this, either show that both sides are true, or that both sides are false, for exactly the same combinations of truth values of the propositional variables in these expressions (whichever is easier).
20. Show that $p \leftrightarrow q$ and $(p \wedge q) \vee (\neg p \wedge \neg q)$ are logically equivalent.
21. Show that $\neg(p \leftrightarrow q)$ and $p \leftrightarrow \neg q$ are logically equivalent.
22. Show that $p \rightarrow q$ and $\neg q \rightarrow \neg p$ are logically equivalent.
23. Show that $\neg p \leftrightarrow q$ and $p \leftrightarrow \neg q$ are logically equivalent.
24. Show that $\neg(p \oplus q)$ and $p \leftrightarrow q$ are logically equivalent.
25. Show that $\neg(p \leftrightarrow q)$ and $\neg p \leftrightarrow q$ are logically equivalent.
26. Show that $(p \rightarrow q) \wedge (p \rightarrow r)$ and $p \rightarrow (q \wedge r)$ are logically equivalent.
27. Show that $(p \rightarrow r) \wedge (q \rightarrow r)$ and $(p \vee q) \rightarrow r$ are logically equivalent.
28. Show that $(p \rightarrow q) \vee (p \rightarrow r)$ and $p \rightarrow (q \vee r)$ are logically equivalent.
29. Show that $(p \rightarrow r) \vee (q \rightarrow r)$ and $(p \wedge q) \rightarrow r$ are logically equivalent.
30. Show that $\neg p \rightarrow (q \rightarrow r)$ and $q \rightarrow (p \vee r)$ are logically equivalent.
31. Show that $p \leftrightarrow q$ and $(p \rightarrow q) \wedge (q \rightarrow p)$ are logically equivalent.
32. Show that $p \leftrightarrow q$ and $\neg p \leftrightarrow \neg q$ are logically equivalent.
33. Show that $(p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$ is a tautology.
-  34. Show that $(p \vee q) \wedge (\neg p \vee r) \rightarrow (q \vee r)$ is a tautology.
35. Show that $(p \rightarrow q) \rightarrow r$ and $p \rightarrow (q \rightarrow r)$ are not logically equivalent.
36. Show that $(p \wedge q) \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$ are not logically equivalent.

37. Show that $(p \rightarrow q) \rightarrow (r \rightarrow s)$ and $(p \rightarrow r) \rightarrow (q \rightarrow s)$ are not logically equivalent.

The **dual** of a compound proposition that contains only the logical operators $\vee$, $\wedge$, and $\neg$ is the compound proposition obtained by replacing each $\vee$ by $\wedge$, each $\wedge$ by $\vee$, each **T** by **F**, and each **F** by **T**. The dual of s is denoted by s^* .

38. Find the dual of each of these compound propositions.

- a) $p \vee \neg q$ b) $p \wedge (q \vee (r \wedge \mathbf{T}))$
c) $(p \wedge \neg q) \vee (q \wedge \mathbf{F})$

39. Find the dual of each of these compound propositions.

- a) $p \wedge \neg q \wedge \neg r$ b) $(p \wedge q \wedge r) \vee s$
c) $(p \vee \mathbf{F}) \wedge (q \vee \mathbf{T})$

40. When does $s^* = s$, where s is a compound proposition?

41. Show that $(s^*)^* = s$ when s is a compound proposition.

42. Show that the logical equivalences in Table 6, except for the double negation law, come in pairs, where each pair contains compound propositions that are duals of each other.

- **43. Why are the duals of two equivalent compound propositions also equivalent, where these compound propositions contain only the operators $\wedge$, $\vee$, and $\neg$?

44. Find a compound proposition involving the propositional variables p , q , and r that is true when p and q are true and r is false, but is false otherwise. [Hint: Use a conjunction of each propositional variable or its negation.]

45. Find a compound proposition involving the propositional variables p , q , and r that is true when exactly two of p , q , and r are true and is false otherwise. [Hint: Form a disjunction of conjunctions. Include a conjunction for each combination of values for which the compound proposition is true. Each conjunction should include each of the three propositional variables or its negations.]

-  46. Suppose that a truth table in n propositional variables is specified. Show that a compound proposition with this truth table can be formed by taking the disjunction of conjunctions of the variables or their negations, with one conjunction included for each combination of values for which the compound proposition is true. The resulting compound proposition is said to be in **disjunctive normal form**.

A collection of logical operators is called **functionally complete** if every compound proposition is logically equivalent to a compound proposition involving only these logical operators.

47. Show that $\neg$, $\wedge$, and $\vee$ form a functionally complete collection of logical operators. [Hint: Use the fact that every compound proposition is logically equivalent to one in disjunctive normal form, as shown in Exercise 46.]

- *48. Show that $\neg$ and $\wedge$ form a functionally complete collection of logical operators. [Hint: First use a De Morgan law to show that $p \vee q$ is logically equivalent to $\neg(\neg p \wedge \neg q)$.]

- *49. Show that $\neg$ and $\vee$ form a functionally complete collection of logical operators.

We now present a group of exercises that involve the logical operators *NAND* and *NOR*. The proposition p *NAND* q is true

when either p or q , or both, are false; and it is false when both p and q are true. The proposition p *NOR* q is true when both p and q are false, and it is false otherwise. The propositions p *NAND* q and p *NOR* q are denoted by $p \mid q$ and $p \downarrow q$, respectively. (The operators $\mid$ and $\downarrow$ are called the **Sheffer stroke** and the **Peirce arrow** after H. M. Sheffer and C. S. Peirce, respectively.)

50. Construct a truth table for the logical operator *NAND*.

51. Show that $p \mid q$ is logically equivalent to $\neg(p \wedge q)$.

52. Construct a truth table for the logical operator *NOR*.

53. Show that $p \downarrow q$ is logically equivalent to $\neg(p \vee q)$.

54. In this exercise we will show that $\{\downarrow\}$ is a functionally complete collection of logical operators.

- a) Show that $p \downarrow p$ is logically equivalent to $\neg p$.

- b) Show that $(p \downarrow q) \downarrow (p \downarrow q)$ is logically equivalent to $p \vee q$.

- c) Conclude from parts (a) and (b), and Exercise 49, that $\{\downarrow\}$ is a functionally complete collection of logical operators.

- *55. Find a compound proposition logically equivalent to $p \rightarrow q$ using only the logical operator $\downarrow$.

56. Show that $\{\mid\}$ is a functionally complete collection of logical operators.

57. Show that $p \mid q$ and $q \mid p$ are equivalent.

58. Show that $p \mid (q \mid r)$ and $(p \mid q) \mid r$ are not equivalent, so that the logical operator $\mid$ is not associative.

- *59. How many different truth tables of compound propositions are there that involve the propositional variables p and q ?

60. Show that if p , q , and r are compound propositions such that p and q are logically equivalent and q and r are logically equivalent, then p and r are logically equivalent.

61. The following sentence is taken from the specification of a telephone system: "If the directory database is opened, then the monitor is put in a closed state, if the system is not in its initial state." This specification is hard to understand because it involves two conditional statements. Find an equivalent, easier-to-understand specification that involves disjunctions and negations but not conditional statements.

62. How many of the disjunctions $p \vee \neg q$, $\neg p \vee q$, $q \vee r$, $q \vee \neg r$, and $\neg q \vee \neg r$ can be made simultaneously true by an assignment of truth values to p , q , and r ?

63. How many of the disjunctions $p \vee \neg q \vee s$, $\neg p \vee \neg r \vee s$, $\neg p \vee \neg r \vee \neg s$, $\neg p \vee q \vee \neg s$, $q \vee r \vee \neg s$, $q \vee \neg r \vee \neg s$, $\neg p \vee \neg q \vee \neg s$, $p \vee r \vee s$, and $p \vee r \vee \neg s$ can be made simultaneously true by an assignment of truth values to p , q , r , and s ?

64. Show that the negation of an unsatisfiable compound proposition is a tautology and the negation of a compound proposition that is a tautology is unsatisfiable.

65. Determine whether each of these compound propositions is satisfiable.

- a) $(p \vee \neg q) \wedge (\neg p \vee q) \wedge (\neg p \vee \neg q)$

- b) $(p \rightarrow q) \wedge (p \rightarrow \neg q) \wedge (\neg p \rightarrow q) \wedge (\neg p \rightarrow \neg q)$

- c) $(p \leftrightarrow q) \wedge (\neg p \leftrightarrow q)$

66. Determine whether each of these compound propositions is satisfiable.

- a) $(p \vee q \vee \neg r) \wedge (p \vee \neg q \vee \neg s) \wedge (p \vee \neg r \vee \neg s) \wedge (\neg p \vee \neg q \vee \neg s) \wedge (p \vee q \vee \neg s)$
 b) $(\neg p \vee \neg q \vee r) \wedge (\neg p \vee q \vee \neg s) \wedge (p \vee \neg q \vee \neg s) \wedge (\neg p \vee \neg r \vee \neg s) \wedge (p \vee q \vee \neg r) \wedge (p \vee \neg r \vee \neg s)$
 c) $(p \vee q \vee r) \wedge (p \vee \neg q \vee \neg s) \wedge (q \vee \neg r \vee s) \wedge (\neg p \vee r \vee s) \wedge (\neg p \vee q \vee \neg s) \wedge (p \vee \neg q \vee \neg r) \wedge (\neg p \vee \neg q \vee s) \wedge (\neg p \vee \neg r \vee \neg s)$

67. Find the compound proposition Q constructed in Example 10 for the n -queens problem, and use it to find all the ways that n queens can be placed on an $n \times n$ chessboard, so that no queen can attack another when n is

- a) 2. b) 3. c) 4.

68. Starting with the compound proposition Q found in Example 10, construct a compound proposition that can be

used to find all solutions of the n -queens problem where the queen in the first column is in an odd-numbered row.

69. Show how the solution of a given 4×4 Sudoku puzzle can be found by solving a satisfiability problem.

70. Construct a compound proposition that asserts that every cell of a 9×9 Sudoku puzzle contains at least one number.

71. Explain the steps in the construction of the compound proposition given in the text that asserts that every column of a 9×9 Sudoku puzzle contains every number.

*72. Explain the steps in the construction of the compound proposition given in the text that asserts that each of the nine 3×3 blocks of a 9×9 Sudoku puzzle contains every number.

1.4 Predicates and Quantifiers

1.4.1 Introduction

Propositional logic, studied in Sections 1.1–1.3, cannot adequately express the meaning of all statements in mathematics and in natural language. For example, suppose that we know that

“Every computer connected to the university network is functioning properly.”

No rules of propositional logic allow us to conclude the truth of the statement

“MATH3 is functioning properly,”

where MATH3 is one of the computers connected to the university network. Likewise, we cannot use the rules of propositional logic to conclude from the statement

“CS2 is under attack by an intruder,”

where CS2 is a computer on the university network, to conclude the truth of

“There is a computer on the university network that is under attack by an intruder.”

In this section we will introduce a more powerful type of logic called **predicate logic**. We will see how predicate logic can be used to express the meaning of a wide range of statements in mathematics and computer science in ways that permit us to reason and explore relationships between objects. To understand predicate logic, we first need to introduce the concept of a predicate. Afterward, we will introduce the notion of quantifiers, which enable us to reason with statements that assert that a certain property holds for all objects of a certain type and with statements that assert the existence of an object with a particular property.

1.4.2 Predicates

Statements involving variables, such as

“ $x > 3$,” “ $x = y + 3$,” “ $x + y = z$,”

Prolog answers queries using the facts and rules it is given. For example, using the facts and rules listed, the query

```
?enrolled(kevin,math273)
```

produces the response

```
yes
```

because the fact *enrolled*(kevin, math273) was provided as input. The query

```
?enrolled(X,math273)
```

produces the response

```
kevin
kiko
```

To produce this response, Prolog determines all possible values of X for which *enrolled*(X , math273) has been included as a Prolog fact. Similarly, to find all the professors who are instructors in classes being taken by Juana, we use the query

```
?teaches(X,juana)
```

This query returns

```
patel
grossman
```



Exercises

- Let $P(x)$ denote the statement “ $x \leq 4$.” What are these truth values?
 - $P(0)$
 - $P(4)$
 - $P(6)$
- Let $P(x)$ be the statement “The word x contains the letter a .” What are these truth values?
 - $P(\text{orange})$
 - $P(\text{lemon})$
 - $P(\text{true})$
 - $P(\text{false})$
- Let $Q(x, y)$ denote the statement “ x is the capital of y .” What are these truth values?
 - $Q(\text{Denver, Colorado})$
 - $Q(\text{Detroit, Michigan})$
 - $Q(\text{Massachusetts, Boston})$
 - $Q(\text{New York, New York})$
- State the value of x after the statement **if** $P(x)$ **then** $x := 1$ is executed, where $P(x)$ is the statement “ $x > 1$,” if the value of x when this statement is reached is
 - $x = 0$.
 - $x = 1$.
 - $x = 2$.
- Let $P(x)$ be the statement “ x spends more than five hours every weekday in class,” where the domain for x consists of all students. Express each of these quantifications in English.
 - $\exists x P(x)$
 - $\forall x P(x)$
 - $\exists x \neg P(x)$
 - $\forall x \neg P(x)$
- Let $N(x)$ be the statement “ x has visited North Dakota,” where the domain consists of the students in your school. Express each of these quantifications in English.
 - $\exists x N(x)$
 - $\forall x N(x)$
 - $\neg \exists x N(x)$
 - $\exists x \neg N(x)$
 - $\neg \forall x N(x)$
 - $\forall x \neg N(x)$
- Translate these statements into English, where $C(x)$ is “ x is a comedian” and $F(x)$ is “ x is funny” and the domain consists of all people.
 - $\forall x (C(x) \rightarrow F(x))$
 - $\forall x (C(x) \wedge F(x))$
 - $\exists x (C(x) \rightarrow F(x))$
 - $\exists x (C(x) \wedge F(x))$
- Translate these statements into English, where $R(x)$ is “ x is a rabbit” and $H(x)$ is “ x hops” and the domain consists of all animals.
 - $\forall x (R(x) \rightarrow H(x))$
 - $\forall x (R(x) \wedge H(x))$
 - $\exists x (R(x) \rightarrow H(x))$
 - $\exists x (R(x) \wedge H(x))$
- Let $P(x)$ be the statement “ x can speak Russian” and let $Q(x)$ be the statement “ x knows the computer language C++.” Express each of these sentences in terms of $P(x)$, $Q(x)$, quantifiers, and logical connectives. The domain for quantifiers consists of all students at your school.

- a) There is a student at your school who can speak Russian and who knows C++.
- b) There is a student at your school who can speak Russian but who doesn't know C++.
- c) Every student at your school either can speak Russian or knows C++.
- d) No student at your school can speak Russian or knows C++.
10. Let $C(x)$ be the statement "x has a cat," let $D(x)$ be the statement "x has a dog," and let $F(x)$ be the statement "x has a ferret." Express each of these statements in terms of $C(x)$, $D(x)$, $F(x)$, quantifiers, and logical connectives. Let the domain consist of all students in your class.
- a) A student in your class has a cat, a dog, and a ferret.
- b) All students in your class have a cat, a dog, or a ferret.
- c) Some student in your class has a cat and a ferret, but not a dog.
- d) No student in your class has a cat, a dog, and a ferret.
- e) For each of the three animals, cats, dogs, and ferrets, there is a student in your class who has this animal as a pet.
11. Let $P(x)$ be the statement " $x = x^2$." If the domain consists of the integers, what are these truth values?
- a) $P(0)$ b) $P(1)$ c) $P(2)$
d) $P(-1)$ e) $\exists xP(x)$ f) $\forall xP(x)$
12. Let $Q(x)$ be the statement " $x + 1 > 2x$." If the domain consists of all integers, what are these truth values?
- a) $Q(0)$ b) $Q(-1)$ c) $Q(1)$
d) $\exists xQ(x)$ e) $\forall xQ(x)$ f) $\exists x\neg Q(x)$
g) $\forall x\neg Q(x)$
13. Determine the truth value of each of these statements if the domain consists of all integers.
- a) $\forall n(n + 1 > n)$ b) $\exists n(2n = 3n)$
c) $\exists n(n = -n)$ d) $\forall n(3n \leq 4n)$
14. Determine the truth value of each of these statements if the domain consists of all real numbers.
- a) $\exists x(x^3 = -1)$ b) $\exists x(x^4 < x^2)$
c) $\forall x((-x)^2 = x^2)$ d) $\forall x(2x > x)$
15. Determine the truth value of each of these statements if the domain for all variables consists of all integers.
- a) $\forall n(n^2 \geq 0)$ b) $\exists n(n^2 = 2)$
c) $\forall n(n^2 \geq n)$ d) $\exists n(n^2 < 0)$
16. Determine the truth value of each of these statements if the domain of each variable consists of all real numbers.
- a) $\exists x(x^2 = 2)$ b) $\exists x(x^2 = -1)$
c) $\forall x(x^2 + 2 \geq 1)$ d) $\forall x(x^2 \neq x)$
17. Suppose that the domain of the propositional function $P(x)$ consists of the integers 0, 1, 2, 3, and 4. Write out each of these propositions using disjunctions, conjunctions, and negations.
- a) $\exists xP(x)$ b) $\forall xP(x)$ c) $\exists x\neg P(x)$
d) $\forall x\neg P(x)$ e) $\neg\exists xP(x)$ f) $\neg\forall xP(x)$
18. Suppose that the domain of the propositional function $P(x)$ consists of the integers $-2, -1, 0, 1$, and 2 . Write out each of these propositions using disjunctions, conjunctions, and negations.
- a) $\exists xP(x)$ b) $\forall xP(x)$ c) $\exists x\neg P(x)$
d) $\forall x\neg P(x)$ e) $\neg\exists xP(x)$ f) $\neg\forall xP(x)$
19. Suppose that the domain of the propositional function $P(x)$ consists of the integers 1, 2, 3, 4, and 5. Express these statements without using quantifiers, instead using only negations, disjunctions, and conjunctions.
- a) $\exists xP(x)$ b) $\forall xP(x)$
c) $\neg\exists xP(x)$ d) $\neg\forall xP(x)$
e) $\forall x((x \neq 3) \rightarrow P(x)) \vee \exists x\neg P(x)$
20. Suppose that the domain of the propositional function $P(x)$ consists of $-5, -3, -1, 1, 3$, and 5 . Express these statements without using quantifiers, instead using only negations, disjunctions, and conjunctions.
- a) $\exists xP(x)$ b) $\forall xP(x)$
c) $\forall x((x \neq 1) \rightarrow P(x))$
d) $\exists x((x \geq 0) \wedge P(x))$
e) $\exists x(\neg P(x)) \wedge \forall x((x < 0) \rightarrow P(x))$
21. For each of these statements find a domain for which the statement is true and a domain for which the statement is false.
- a) Everyone is studying discrete mathematics.
b) Everyone is older than 21 years.
c) Every two people have the same mother.
d) No two different people have the same grandmother.
22. For each of these statements find a domain for which the statement is true and a domain for which the statement is false.
- a) Everyone speaks Hindi.
b) There is someone older than 21 years.
c) Every two people have the same first name.
d) Someone knows more than two other people.
23. Translate in two ways each of these statements into logical expressions using predicates, quantifiers, and logical connectives. First, let the domain consist of the students in your class and second, let it consist of all people.
- a) Someone in your class can speak Hindi.
b) Everyone in your class is friendly.
c) There is a person in your class who was not born in California.
d) A student in your class has been in a movie.
e) No student in your class has taken a course in logic programming.
24. Translate in two ways each of these statements into logical expressions using predicates, quantifiers, and logical connectives. First, let the domain consist of the students in your class and second, let it consist of all people.
- a) Everyone in your class has a cellular phone.
b) Somebody in your class has seen a foreign movie.
c) There is a person in your class who cannot swim.
d) All students in your class can solve quadratic equations.
e) Some student in your class does not want to be rich.
25. Translate each of these statements into logical expressions using predicates, quantifiers, and logical connectives.

- a) No one is perfect.
 b) Not everyone is perfect.
 c) All your friends are perfect.
 d) At least one of your friends is perfect.
 e) Everyone is your friend and is perfect.
 f) Not everybody is your friend or someone is not perfect.
26. Translate each of these statements into logical expressions in three different ways by varying the domain and by using predicates with one and with two variables.
 a) Someone in your school has visited Uzbekistan.
 b) Everyone in your class has studied calculus and C++.
 c) No one in your school owns both a bicycle and a motorcycle.
 d) There is a person in your school who is not happy.
 e) Everyone in your school was born in the twentieth century.
27. Translate each of these statements into logical expressions in three different ways by varying the domain and by using predicates with one and with two variables.
 a) A student in your school has lived in Vietnam.
 b) There is a student in your school who cannot speak Hindi.
 c) A student in your school knows Java, Prolog, and C++.
 d) Everyone in your class enjoys Thai food.
 e) Someone in your class does not play hockey.
28. Translate each of these statements into logical expressions using predicates, quantifiers, and logical connectives.
 a) Something is not in the correct place.
 b) All tools are in the correct place and are in excellent condition.
 c) Everything is in the correct place and in excellent condition.
 d) Nothing is in the correct place and is in excellent condition.
 e) One of your tools is not in the correct place, but it is in excellent condition.
29. Express each of these statements using logical operators, predicates, and quantifiers.
 a) Some propositions are tautologies.
 b) The negation of a contradiction is a tautology.
 c) The disjunction of two contingencies can be a tautology.
 d) The conjunction of two tautologies is a tautology.
30. Suppose the domain of the propositional function $P(x, y)$ consists of pairs x and y , where x is 1, 2, or 3 and y is 1, 2, or 3. Write out these propositions using disjunctions and conjunctions.
- a) $\exists x P(x, 3)$
 b) $\forall y P(1, y)$
 c) $\exists y \neg P(2, y)$
 d) $\forall x \neg P(x, 2)$
31. Suppose that the domain of $Q(x, y, z)$ consists of triples x, y, z , where $x = 0, 1$, or 2 , $y = 0$ or 1 , and $z = 0$ or 1 . Write out these propositions using disjunctions and conjunctions.
 a) $\forall y Q(0, y, 0)$
 b) $\exists x Q(x, 1, 1)$
 c) $\exists z \neg Q(0, 0, z)$
 d) $\exists x \neg Q(x, 0, 1)$
32. Express each of these statements using quantifiers. Then form the negation of the statement so that no negation is to the left of a quantifier. Next, express the negation in simple English. (Do not simply use the phrase “It is not the case that.”)
 a) All dogs have fleas.
 b) There is a horse that can add.
 c) Every koala can climb.
 d) No monkey can speak French.
 e) There exists a pig that can swim and catch fish.
33. Express each of these statements using quantifiers. Then form the negation of the statement, so that no negation is to the left of a quantifier. Next, express the negation in simple English. (Do not simply use the phrase “It is not the case that.”)
 a) Some old dogs can learn new tricks.
 b) No rabbit knows calculus.
 c) Every bird can fly.
 d) There is no dog that can talk.
 e) There is no one in this class who knows French and Russian.
34. Express the negation of these propositions using quantifiers, and then express the negation in English.
 a) Some drivers do not obey the speed limit.
 b) All Swedish movies are serious.
 c) No one can keep a secret.
 d) There is someone in this class who does not have a good attitude.
35. Express the negation of each of these statements in terms of quantifiers without using the negation symbol.
 a) $\forall x (x > 1)$
 b) $\forall x (x \leq 2)$
 c) $\exists x (x \geq 4)$
 d) $\exists x (x < 0)$
 e) $\forall x ((x < -1) \vee (x > 2))$
 f) $\exists x ((x < 4) \vee (x > 7))$
36. Express the negation of each of these statements in terms of quantifiers without using the negation symbol.
 a) $\forall x (-2 < x < 3)$
 b) $\forall x (0 \leq x < 5)$
 c) $\exists x (-4 \leq x \leq 1)$
 d) $\exists x (-5 < x < -1)$
37. Find a counterexample, if possible, to these universally quantified statements, where the domain for all variables consists of all integers.
 a) $\forall x (x^2 \geq x)$
 b) $\forall x (x > 0 \vee x < 0)$
 c) $\forall x (x = 1)$

38. Find a counterexample, if possible, to these universally quantified statements, where the domain for all variables consists of all real numbers.

- a) $\forall x(x^2 \neq x)$ b) $\forall x(x^2 \neq 2)$
c) $\forall x(|x| > 0)$

39. Express each of these statements using predicates and quantifiers.

- a) A passenger on an airline qualifies as an elite flyer if the passenger flies more than 25,000 miles in a year or takes more than 25 flights during that year.
b) A man qualifies for the marathon if his best previous time is less than 3 hours and a woman qualifies for the marathon if her best previous time is less than 3.5 hours.
c) A student must take at least 60 course hours, or at least 45 course hours and write a master's thesis, and receive a grade no lower than a B in all required courses, to receive a master's degree.
d) There is a student who has taken more than 21 credit hours in a semester and received all A's.

Exercises 40–44 deal with the translation between system specification and logical expressions involving quantifiers.

40. Translate these system specifications into English, where the predicate $S(x, y)$ is “ x is in state y ” and where the domain for x and y consists of all systems and all possible states, respectively.

- a) $\exists xS(x, \text{open})$
b) $\forall x(S(x, \text{malfunctioning}) \vee S(x, \text{diagnostic}))$
c) $\exists xS(x, \text{open}) \vee \exists xS(x, \text{diagnostic})$
d) $\exists x\neg S(x, \text{available})$
e) $\forall x\neg S(x, \text{working})$

41. Translate these specifications into English, where $F(p)$ is “Printer p is out of service,” $B(p)$ is “Printer p is busy,” $L(j)$ is “Print job j is lost,” and $Q(j)$ is “Print job j is queued.”

- a) $\exists p(F(p) \wedge B(p)) \rightarrow \exists jL(j)$
b) $\forall pB(p) \rightarrow \exists jQ(j)$
c) $\exists j(Q(j) \wedge L(j)) \rightarrow \exists pF(p)$
d) $(\forall pB(p) \wedge \forall jQ(j)) \rightarrow \exists jL(j)$

42. Express each of these system specifications using predicates, quantifiers, and logical connectives.

- a) When there is less than 30 megabytes free on the hard disk, a warning message is sent to all users.
b) No directories in the file system can be opened and no files can be closed when system errors have been detected.
c) The file system cannot be backed up if there is a user currently logged on.
d) Video on demand can be delivered when there are at least 8 megabytes of memory available and the connection speed is at least 56 kilobits per second.

43. Express each of these system specifications using predicates, quantifiers, and logical connectives.

- a) At least one mail message, among the nonempty set of messages, can be saved if there is a disk with more than 10 kilobytes of free space.

- b) Whenever there is an active alert, all queued messages are transmitted.
c) The diagnostic monitor tracks the status of all systems except the main console.
d) Each participant on the conference call whom the host of the call did not put on a special list was billed.

44. Express each of these system specifications using predicates, quantifiers, and logical connectives.

- a) Every user has access to an electronic mailbox.
b) The system mailbox can be accessed by everyone in the group if the file system is locked.
c) The firewall is in a diagnostic state only if the proxy server is in a diagnostic state.
d) At least one router is functioning normally if the throughput is between 100 kbps and 500 kbps and the proxy server is not in diagnostic mode.

45. Determine whether $\forall x(P(x) \rightarrow Q(x))$ and $\forall xP(x) \rightarrow \forall xQ(x)$ are logically equivalent. Justify your answer.

46. Determine whether $\forall x(P(x) \leftrightarrow Q(x))$ and $\forall x P(x) \leftrightarrow \forall xQ(x)$ are logically equivalent. Justify your answer.

47. Show that $\exists x(P(x) \vee Q(x))$ and $\exists xP(x) \vee \exists xQ(x)$ are logically equivalent.

Exercises 48–51 establish rules for **null quantification** that we can use when a quantified variable does not appear in part of a statement.

48. Establish these logical equivalences, where x does not occur as a free variable in A . Assume that the domain is nonempty.

- a) $(\forall xP(x)) \vee A \equiv \forall x(P(x) \vee A)$
b) $(\exists xP(x)) \vee A \equiv \exists x(P(x) \vee A)$

49. Establish these logical equivalences, where x does not occur as a free variable in A . Assume that the domain is nonempty.

- a) $(\forall xP(x)) \wedge A \equiv \forall x(P(x) \wedge A)$
b) $(\exists xP(x)) \wedge A \equiv \exists x(P(x) \wedge A)$

50. Establish these logical equivalences, where x does not occur as a free variable in A . Assume that the domain is nonempty.

- a) $\forall x(A \rightarrow P(x)) \equiv A \rightarrow \forall xP(x)$
b) $\exists x(A \rightarrow P(x)) \equiv A \rightarrow \exists xP(x)$

51. Establish these logical equivalences, where x does not occur as a free variable in A . Assume that the domain is nonempty.

- a) $\forall x(P(x) \rightarrow A) \equiv \exists xP(x) \rightarrow A$
b) $\exists x(P(x) \rightarrow A) \equiv \forall xP(x) \rightarrow A$

52. Show that $\forall xP(x) \vee \forall xQ(x)$ and $\forall x(P(x) \vee Q(x))$ are not logically equivalent.

53. Show that $\exists xP(x) \wedge \exists xQ(x)$ and $\exists x(P(x) \wedge Q(x))$ are not logically equivalent.

54. As mentioned in the text, the notation $\exists!xP(x)$ denotes

“There exists a unique x such that $P(x)$ is true.”

If the domain consists of all integers, what are the truth values of these statements?

- a) $\exists!x(x > 1)$ b) $\exists!x(x^2 = 1)$
c) $\exists!x(x + 3 = 2x)$ d) $\exists!x(x = x + 1)$

55. What are the truth values of these statements?

- a) $\exists!xP(x) \rightarrow \exists xP(x)$
- b) $\forall xP(x) \rightarrow \exists!xP(x)$
- c) $\exists!x\neg P(x) \rightarrow \neg\forall xP(x)$

56. Write out $\exists!xP(x)$, where the domain consists of the integers 1, 2, and 3, in terms of negations, conjunctions, and disjunctions.

57. Given the Prolog facts in Example 28, what would Prolog return given these queries?

- a) `?instructor(chan,math273)`
- b) `?instructor(patel,cs301)`
- c) `?enrolled(X,cs301)`
- d) `?enrolled(kiko,Y)`
- e) `?teaches(grossman,Y)`

58. Given the Prolog facts in Example 28, what would Prolog return when given these queries?

- a) `?enrolled(kevin,ee222)`
- b) `?enrolled(kiko,math273)`
- c) `?instructor(grossman,X)`
- d) `?instructor(X,cs301)`
- e) `?teaches(X,kevin)`

59. Suppose that Prolog facts are used to define the predicates *mother*(*M*, *Y*) and *father*(*F*, *X*), which represent that *M* is the mother of *Y* and *F* is the father of *X*, respectively. Give a Prolog rule to define the predicate *sibling*(*X*, *Y*), which represents that *X* and *Y* are siblings (that is, have the same mother and the same father).

60. Suppose that Prolog facts are used to define the predicates *mother*(*M*, *Y*) and *father*(*F*, *X*), which represent that *M* is the mother of *Y* and *F* is the father of *X*, respectively. Give a Prolog rule to define the predicate *grandfather*(*X*, *Y*), which represents that *X* is the grandfather of *Y*. [Hint: You can write a disjunction in Prolog either by using a semicolon to separate predicates or by putting these predicates on separate lines.]

Exercises 61–64 are based on questions found in the book *Symbolic Logic* by Lewis Carroll.

61. Let *P*(*x*), *Q*(*x*), and *R*(*x*) be the statements “*x* is a professor,” “*x* is ignorant,” and “*x* is vain,” respectively. Express each of these statements using quantifiers; logical connectives; and *P*(*x*), *Q*(*x*), and *R*(*x*), where the domain consists of all people.

a) No professors are ignorant.

b) All ignorant people are vain.

c) No professors are vain.

d) Does (c) follow from (a) and (b)?

62. Let *P*(*x*), *Q*(*x*), and *R*(*x*) be the statements “*x* is a clear explanation,” “*x* is satisfactory,” and “*x* is an excuse,” respectively. Suppose that the domain for *x* consists of all English text. Express each of these statements using quantifiers, logical connectives, and *P*(*x*), *Q*(*x*), and *R*(*x*).

a) All clear explanations are satisfactory.

b) Some excuses are unsatisfactory.

c) Some excuses are not clear explanations.

*d) Does (c) follow from (a) and (b)?

63. Let *P*(*x*), *Q*(*x*), *R*(*x*), and *S*(*x*) be the statements “*x* is a baby,” “*x* is logical,” “*x* is able to manage a crocodile,” and “*x* is despised,” respectively. Suppose that the domain consists of all people. Express each of these statements using quantifiers; logical connectives; and *P*(*x*), *Q*(*x*), *R*(*x*), and *S*(*x*).

a) Babies are illogical.

b) Nobody is despised who can manage a crocodile.

c) Illogical persons are despised.

d) Babies cannot manage crocodiles.

*e) Does (d) follow from (a), (b), and (c)? If not, is there a correct conclusion?

64. Let *P*(*x*), *Q*(*x*), *R*(*x*), and *S*(*x*) be the statements “*x* is a duck,” “*x* is one of my poultry,” “*x* is an officer,” and “*x* is willing to waltz,” respectively. Express each of these statements using quantifiers; logical connectives; and *P*(*x*), *Q*(*x*), *R*(*x*), and *S*(*x*).

a) No ducks are willing to waltz.

b) No officers ever decline to waltz.

c) All my poultry are ducks.

d) My poultry are not officers.

*e) Does (d) follow from (a), (b), and (c)? If not, is there a correct conclusion?

1.5 Nested Quantifiers

1.5.1 Introduction

In Section 1.4 we defined the existential and universal quantifiers and showed how they can be used to represent mathematical statements. We also explained how they can be used to translate English sentences into logical expressions. However, in Section 1.4 we avoided **nested quantifiers**, where one quantifier is within the scope of another, such as

$$\forall x\exists y(x + y = 0).$$

Successively applying the rules for negating quantified expressions, we construct this sequence of equivalent statements:

$$\begin{aligned}
 & \neg \forall \epsilon > 0 \exists \delta > 0 \forall x (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon) \\
 & \equiv \exists \epsilon > 0 \neg \exists \delta > 0 \forall x (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon) \\
 & \equiv \exists \epsilon > 0 \forall \delta > 0 \neg \forall x (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon) \\
 & \equiv \exists \epsilon > 0 \forall \delta > 0 \exists x \neg (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon) \\
 & \equiv \exists \epsilon > 0 \forall \delta > 0 \exists x (0 < |x - a| < \delta \wedge |f(x) - L| \geq \epsilon).
 \end{aligned}$$

In the last step we used the equivalence $\neg(p \rightarrow q) \equiv p \wedge \neg q$, which follows from the fifth equivalence in Table 7 of Section 1.3.

Because the statement “ $\lim_{x \rightarrow a} f(x)$ does not exist” means for all real numbers L , $\lim_{x \rightarrow a} f(x) \neq L$, this can be expressed as

$$\forall L \exists \epsilon > 0 \forall \delta > 0 \exists x (0 < |x - a| < \delta \wedge |f(x) - L| \geq \epsilon).$$

This last statement says that for every real number L there is a real number $\epsilon > 0$ such that for every real number $\delta > 0$, there exists a real number x such that $0 < |x - a| < \delta$ and $|f(x) - L| \geq \epsilon$. ◀

Exercises

- Translate these statements into English, where the domain for each variable consists of all real numbers.
 - $\forall x \exists y (x < y)$
 - $\forall x \forall y (((x \geq 0) \wedge (y \geq 0)) \rightarrow (xy \geq 0))$
 - $\forall x \forall y \exists z (xy = z)$
- Translate these statements into English, where the domain for each variable consists of all real numbers.
 - $\exists x \forall y (xy = y)$
 - $\forall x \forall y (((x \geq 0) \wedge (y < 0)) \rightarrow (x - y > 0))$
 - $\forall x \forall y \exists z (x = y + z)$
- Let $Q(x, y)$ be the statement “ x has sent an e-mail message to y ,” where the domain for both x and y consists of all students in your class. Express each of these quantifications in English.

a) $\exists x \exists y Q(x, y)$	b) $\exists x \forall y Q(x, y)$
c) $\forall x \exists y Q(x, y)$	d) $\exists y \forall x Q(x, y)$
e) $\forall y \exists x Q(x, y)$	f) $\forall x \forall y Q(x, y)$
- Let $P(x, y)$ be the statement “Student x has taken class y ,” where the domain for x consists of all students in your class and for y consists of all computer science courses at your school. Express each of these quantifications in English.

a) $\exists x \exists y P(x, y)$	b) $\exists x \forall y P(x, y)$
c) $\forall x \exists y P(x, y)$	d) $\exists y \forall x P(x, y)$
e) $\forall y \exists x P(x, y)$	f) $\forall x \forall y P(x, y)$
- Let $W(x, y)$ mean that student x has visited website y , where the domain for x consists of all students in your school and the domain for y consists of all websites. Express each of these statements by a simple English sentence.
 - $W(\text{Sarah Smith}, \text{www.att.com})$
 - $\exists x W(x, \text{www.imdb.org})$
 - $\exists y W(\text{José Orez}, y)$
 - $\exists y (W(\text{Ashok Puri}, y) \wedge W(\text{Cindy Yoon}, y))$
 - $\exists y \forall z (y \neq (\text{David Belcher}) \wedge (W(\text{David Belcher}, z) \rightarrow W(y, z)))$
 - $\exists x \exists y \forall z ((x \neq y) \wedge (W(x, z) \leftrightarrow W(y, z)))$
- Let $C(x, y)$ mean that student x is enrolled in class y , where the domain for x consists of all students in your school and the domain for y consists of all classes being given at your school. Express each of these statements by a simple English sentence.
 - $C(\text{Randy Goldberg}, \text{CS 252})$
 - $\exists x C(x, \text{Math 695})$
 - $\exists y C(\text{Carol Sitea}, y)$
 - $\exists x (C(x, \text{Math 222}) \wedge C(x, \text{CS 252}))$
 - $\exists x \exists y \forall z ((x \neq y) \wedge (C(x, z) \rightarrow C(y, z)))$
 - $\exists x \exists y \forall z ((x \neq y) \wedge (C(x, z) \leftrightarrow C(y, z)))$
- Let $T(x, y)$ mean that student x likes cuisine y , where the domain for x consists of all students at your school and the domain for y consists of all cuisines. Express each of these statements by a simple English sentence.
 - $\neg T(\text{Abdallah Hussein}, \text{Japanese})$

- b) $\exists x T(x, \text{Korean}) \wedge \forall x T(x, \text{Mexican})$
 c) $\exists y (T(\text{Monique Arsenault}, y) \vee T(\text{Jay Johnson}, y))$
 d) $\forall x \forall z \exists y ((x \neq z) \rightarrow \neg(T(x, y) \wedge T(z, y)))$
 e) $\exists x \exists z \forall y (T(x, y) \leftrightarrow T(z, y))$
 f) $\forall x \forall z \exists y (T(x, y) \leftrightarrow T(z, y))$
8. Let $Q(x, y)$ be the statement “Student x has been a contestant on quiz show y .” Express each of these sentences in terms of $Q(x, y)$, quantifiers, and logical connectives, where the domain for x consists of all students at your school and for y consists of all quiz shows on television.
- There is a student at your school who has been a contestant on a television quiz show.
 - No student at your school has ever been a contestant on a television quiz show.
 - There is a student at your school who has been a contestant on *Jeopardy!* and on *Wheel of Fortune*.
 - Every television quiz show has had a student from your school as a contestant.
 - At least two students from your school have been contestants on *Jeopardy!*.
9. Let $L(x, y)$ be the statement “ x loves y ,” where the domain for both x and y consists of all people in the world. Use quantifiers to express each of these statements.
- Everybody loves Jerry.
 - Everybody loves somebody.
 - There is somebody whom everybody loves.
 - Nobody loves everybody.
 - There is somebody whom Lydia does not love.
 - There is somebody whom no one loves.
 - There is exactly one person whom everybody loves.
 - There are exactly two people whom Lynn loves.
 - Everyone loves himself or herself.
 - There is someone who loves no one besides himself or herself.
10. Let $F(x, y)$ be the statement “ x can fool y ,” where the domain consists of all people in the world. Use quantifiers to express each of these statements.
- Everybody can fool Fred.
 - Evelyn can fool everybody.
 - Everybody can fool somebody.
 - There is no one who can fool everybody.
 - Everyone can be fooled by somebody.
 - No one can fool both Fred and Jerry.
 - Nancy can fool exactly two people.
 - There is exactly one person whom everybody can fool.
 - No one can fool himself or herself.
 - There is someone who can fool exactly one person besides himself or herself.
11. Let $S(x)$ be the predicate “ x is a student,” $F(x)$ the predicate “ x is a faculty member,” and $A(x, y)$ the predicate “ x has asked y a question,” where the domain consists of all people associated with your school. Use quantifiers to express each of these statements.
- Lois has asked Professor Michaels a question.
 - Every student has asked Professor Gross a question.
 - Every faculty member has either asked Professor Miller a question or been asked a question by Professor Miller.
 - Some student has not asked any faculty member a question.
 - There is a faculty member who has never been asked a question by a student.
 - Some student has asked every faculty member a question.
 - There is a faculty member who has asked every other faculty member a question.
 - Some student has never been asked a question by a faculty member.
12. Let $I(x)$ be the statement “ x has an Internet connection” and $C(x, y)$ be the statement “ x and y have chatted over the Internet,” where the domain for the variables x and y consists of all students in your class. Use quantifiers to express each of these statements.
- Jerry does not have an Internet connection.
 - Rachel has not chatted over the Internet with Chelsea.
 - Jan and Sharon have never chatted over the Internet.
 - No one in the class has chatted with Bob.
 - Sanjay has chatted with everyone except Joseph.
 - Someone in your class does not have an Internet connection.
 - Not everyone in your class has an Internet connection.
 - Exactly one student in your class has an Internet connection.
 - Everyone except one student in your class has an Internet connection.
 - Everyone in your class with an Internet connection has chatted over the Internet with at least one other student in your class.
 - Someone in your class has an Internet connection but has not chatted with anyone else in your class.
 - There are two students in your class who have not chatted with each other over the Internet.
 - There is a student in your class who has chatted with everyone in your class over the Internet.
 - There are at least two students in your class who have not chatted with the same person in your class.
 - There are two students in the class who between them have chatted with everyone else in the class.
13. Let $M(x, y)$ be “ x has sent y an e-mail message” and $T(x, y)$ be “ x has telephoned y ,” where the domain consists of all students in your class. Use quantifiers to express each of these statements. (Assume that all e-mail messages that were sent are received, which is not the way things often work.)
- Chou has never sent an e-mail message to Koko.
 - Arlene has never sent an e-mail message to or telephoned Sarah.
 - José has never received an e-mail message from Deborah.
 - Every student in your class has sent an e-mail message to Ken.
 - No one in your class has telephoned Nina.

- f) Everyone in your class has either telephoned Avi or sent him an e-mail message.
 - g) There is a student in your class who has sent everyone else in your class an e-mail message.
 - h) There is someone in your class who has either sent an e-mail message or telephoned everyone else in your class.
 - i) There are two different students in your class who have sent each other e-mail messages.
 - j) There is a student who has sent himself or herself an e-mail message.
 - k) There is a student in your class who has not received an e-mail message from anyone else in the class and who has not been called by any other student in the class.
 - l) Every student in the class has either received an e-mail message or received a telephone call from another student in the class.
 - m) There are at least two students in your class such that one student has sent the other e-mail and the second student has telephoned the first student.
 - n) There are two different students in your class who between them have sent an e-mail message to or telephoned everyone else in the class.
14. Use quantifiers and predicates with more than one variable to express these statements.
- a) There is a student in this class who can speak Hindi.
 - b) Every student in this class plays some sport.
 - c) Some student in this class has visited Alaska but has not visited Hawaii.
 - d) All students in this class have learned at least one programming language.
 - e) There is a student in this class who has taken every course offered by one of the departments in this school.
 - f) Some student in this class grew up in the same town as exactly one other student in this class.
 - g) Every student in this class has chatted with at least one other student in at least one chat group.
15. Use quantifiers and predicates with more than one variable to express these statements.
- a) Every computer science student needs a course in discrete mathematics.
 - b) There is a student in this class who owns a personal computer.
 - c) Every student in this class has taken at least one computer science course.
 - d) There is a student in this class who has taken at least one course in computer science.
 - e) Every student in this class has been in every building on campus.
 - f) There is a student in this class who has been in every room of at least one building on campus.
 - g) Every student in this class has been in at least one room of every building on campus.
16. A discrete mathematics class contains 1 mathematics major who is a freshman, 12 mathematics majors who are sophomores, 15 computer science majors who are sophomores, 2 mathematics majors who are juniors, 2 computer science majors who are juniors, and 1 computer science major who is a senior. Express each of these statements in terms of quantifiers and then determine its truth value.
- a) There is a student in the class who is a junior.
 - b) Every student in the class is a computer science major.
 - c) There is a student in the class who is neither a mathematics major nor a junior.
 - d) Every student in the class is either a sophomore or a computer science major.
 - e) There is a major such that there is a student in the class in every year of study with that major.
17. Express each of these system specifications using predicates, quantifiers, and logical connectives, if necessary.
- a) Every user has access to exactly one mailbox.
 - b) There is a process that continues to run during all error conditions only if the kernel is working correctly.
 - c) All users on the campus network can access all websites whose url has a .edu extension.
 - * d) There are exactly two systems that monitor every remote server.
18. Express each of these system specifications using predicates, quantifiers, and logical connectives, if necessary.
- a) At least one console must be accessible during every fault condition.
 - b) The e-mail address of every user can be retrieved whenever the archive contains at least one message sent by every user on the system.
 - c) For every security breach there is at least one mechanism that can detect that breach if and only if there is a process that has not been compromised.
 - d) There are at least two paths connecting every two distinct endpoints on the network.
 - e) No one knows the password of every user on the system except for the system administrator, who knows all passwords.
19. Express each of these statements using mathematical and logical operators, predicates, and quantifiers, where the domain consists of all integers.
- a) The sum of two negative integers is negative.
 - b) The difference of two positive integers is not necessarily positive.
 - c) The sum of the squares of two integers is greater than or equal to the square of their sum.
 - d) The absolute value of the product of two integers is the product of their absolute values.
20. Express each of these statements using predicates, quantifiers, logical connectives, and mathematical operators where the domain consists of all integers.
- a) The product of two negative integers is positive.
 - b) The average of two positive integers is positive.

- c) The difference of two negative integers is not necessarily negative.
 d) The absolute value of the sum of two integers does not exceed the sum of the absolute values of these integers.
21. Use predicates, quantifiers, logical connectives, and mathematical operators to express the statement that every positive integer is the sum of the squares of four integers.
22. Use predicates, quantifiers, logical connectives, and mathematical operators to express the statement that there is a positive integer that is not the sum of three squares.
23. Express each of these mathematical statements using predicates, quantifiers, logical connectives, and mathematical operators.
- a) The product of two negative real numbers is positive.
 b) The difference of a real number and itself is zero.
 c) Every positive real number has exactly two square roots.
 d) A negative real number does not have a square root that is a real number.
24. Translate each of these nested quantifications into an English statement that expresses a mathematical fact. The domain in each case consists of all real numbers.
- a) $\exists x \forall y (x + y = y)$
 b) $\forall x \forall y ((x \geq 0) \wedge (y < 0)) \rightarrow (x - y > 0)$
 c) $\exists x \exists y ((x \leq 0) \wedge (y \leq 0) \wedge (x - y > 0))$
 d) $\forall x \forall y ((x \neq 0) \wedge (y \neq 0) \leftrightarrow (xy \neq 0))$
25. Translate each of these nested quantifications into an English statement that expresses a mathematical fact. The domain in each case consists of all real numbers.
- a) $\exists x \forall y (xy = y)$
 b) $\forall x \forall y ((x < 0) \wedge (y < 0)) \rightarrow (xy > 0)$
 c) $\exists x \exists y ((x^2 > y) \wedge (x < y))$
 d) $\forall x \forall y \exists z (x + y = z)$
26. Let $Q(x, y)$ be the statement “ $x + y = x - y$.” If the domain for both variables consists of all integers, what are the truth values?
- a) $Q(1, 1)$ b) $Q(2, 0)$
 c) $\forall y Q(1, y)$ d) $\exists x Q(x, 2)$
 e) $\exists x \exists y Q(x, y)$ f) $\forall x \exists y Q(x, y)$
 g) $\exists y \forall x Q(x, y)$ h) $\forall y \exists x Q(x, y)$
 i) $\forall x \forall y Q(x, y)$
27. Determine the truth value of each of these statements if the domain for all variables consists of all integers.
- a) $\forall n \exists m (n^2 < m)$ b) $\exists n \forall m (n < m^2)$
 c) $\forall n \exists m (n + m = 0)$ d) $\exists n \forall m (nm = m)$
 e) $\exists n \exists m (n^2 + m^2 = 5)$ f) $\exists n \exists m (n^2 + m^2 = 6)$
 g) $\exists n \exists m (n + m = 4 \wedge n - m = 1)$
 h) $\exists n \exists m (n + m = 4 \wedge n - m = 2)$
 i) $\forall n \forall m \exists p (p = (m + n)/2)$
28. Determine the truth value of each of these statements if the domain of each variable consists of all real numbers.
- a) $\forall x \exists y (x^2 = y)$ b) $\forall x \exists y (x = y^2)$
 c) $\exists x \forall y (xy = 0)$ d) $\exists x \exists y (x + y \neq y + x)$
- e) $\forall x (x \neq 0 \rightarrow \exists y (xy = 1))$
 f) $\exists x \forall y (y \neq 0 \rightarrow xy = 1)$
 g) $\forall x \exists y (x + y = 1)$
 h) $\exists x \exists y (x + 2y = 2 \wedge 2x + 4y = 5)$
 i) $\forall x \exists y (x + y = 2 \wedge 2x - y = 1)$
 j) $\forall x \forall y \exists z (z = (x + y)/2)$
29. Suppose the domain of the propositional function $P(x, y)$ consists of pairs x and y , where x is 1, 2, or 3 and y is 1, 2, or 3. Write out these propositions using disjunctions and conjunctions.
- a) $\forall x \forall y P(x, y)$ b) $\exists x \exists y P(x, y)$
 c) $\exists x \forall y P(x, y)$ d) $\forall y \exists x P(x, y)$
30. Rewrite each of these statements so that negations appear only within predicates (that is, so that no negation is outside a quantifier or an expression involving logical connectives).
- a) $\neg \exists y \exists x P(x, y)$ b) $\neg \forall x \exists y P(x, y)$
 c) $\neg \exists y (Q(y) \wedge \forall x \neg R(x, y))$
 d) $\neg \exists y (\exists x R(x, y) \vee \forall x S(x, y))$
 e) $\neg \exists y (\forall x \exists z T(x, y, z) \vee \exists x \forall z U(x, y, z))$
31. Express the negations of each of these statements so that all negation symbols immediately precede predicates.
- a) $\forall x \exists y \forall z T(x, y, z)$
 b) $\forall x \exists y P(x, y) \vee \forall x \exists y Q(x, y)$
 c) $\forall x \exists y (P(x, y) \wedge \exists z R(x, y, z))$
 d) $\forall x \exists y (P(x, y) \rightarrow Q(x, y))$
32. Express the negations of each of these statements so that all negation symbols immediately precede predicates.
- a) $\exists z \forall y \forall x T(x, y, z)$
 b) $\exists x \exists y P(x, y) \wedge \forall x \forall y Q(x, y)$
 c) $\exists x \exists y (Q(x, y) \leftrightarrow Q(y, x))$
 d) $\forall y \exists x \exists z (T(x, y, z) \vee Q(x, y))$
33. Rewrite each of these statements so that negations appear only within predicates (that is, so that no negation is outside a quantifier or an expression involving logical connectives).
- a) $\neg \forall x \forall y P(x, y)$ b) $\neg \forall y \exists x P(x, y)$
 c) $\neg \forall y \forall x (P(x, y) \vee Q(x, y))$
 d) $\neg (\exists x \exists y \neg P(x, y) \wedge \forall x \forall y Q(x, y))$
 e) $\neg \forall x (\exists y \forall z P(x, y, z) \wedge \exists z \forall y P(x, y, z))$
34. Find a common domain for the variables x, y , and z for which the statement $\forall x \forall y ((x \neq y) \rightarrow \forall z ((z = x) \vee (z = y)))$ is true and another domain for which it is false.
35. Find a common domain for the variables x, y, z , and w for which the statement $\forall x \forall y \forall z \exists w ((w \neq x) \wedge (w \neq y) \wedge (w \neq z))$ is true and another common domain for these variables for which it is false.
36. Express each of these statements using quantifiers. Then form the negation of the statement so that no negation is to the left of a quantifier. Next, express the negation in simple English. (Do not simply use the phrase “It is not the case that.”)
- a) No one has lost more than one thousand dollars playing the lottery.
 b) There is a student in this class who has chatted with exactly one other student.

- c) No student in this class has sent e-mail to exactly two other students in this class.
 - d) Some student has solved every exercise in this book.
 - e) No student has solved at least one exercise in every section of this book.
37. Express each of these statements using quantifiers. Then form the negation of the statement so that no negation is to the left of a quantifier. Next, express the negation in simple English. (Do not simply use the phrase “It is not the case that.”)
- a) Every student in this class has taken exactly two mathematics classes at this school.
 - b) Someone has visited every country in the world except Libya.
 - c) No one has climbed every mountain in the Himalayas.
 - d) Every movie actor has either been in a movie with Kevin Bacon or has been in a movie with someone who has been in a movie with Kevin Bacon.
38. Express the negations of these propositions using quantifiers, and in English.
- a) Every student in this class likes mathematics.
 - b) There is a student in this class who has never seen a computer.
 - c) There is a student in this class who has taken every mathematics course offered at this school.
 - d) There is a student in this class who has been in at least one room of every building on campus.
39. Find a counterexample, if possible, to these universally quantified statements, where the domain for all variables consists of all integers.
- a) $\forall x \forall y (x^2 = y^2 \rightarrow x = y)$
 - b) $\forall x \exists y (y^2 = x)$
 - c) $\forall x \forall y (xy \geq x)$
40. Find a counterexample, if possible, to these universally quantified statements, where the domain for all variables consists of all integers.
- a) $\forall x \exists y (x = 1/y)$
 - b) $\forall x \exists y (y^2 - x < 100)$
 - c) $\forall x \forall y (x^2 \neq y^3)$
41. Use quantifiers to express the associative law for multiplication of real numbers.
42. Use quantifiers to express the distributive laws of multiplication over addition for real numbers.
43. Use quantifiers and logical connectives to express the fact that every linear polynomial (that is, polynomial of degree 1) with real coefficients and where the coefficient of x is nonzero, has exactly one real root.
44. Use quantifiers and logical connectives to express the fact that a quadratic polynomial with real number coefficients has at most two real roots.
45. Determine the truth value of the statement $\forall x \exists y (xy = 1)$ if the domain for the variables consists of

- a) the nonzero real numbers.
- b) the nonzero integers.
- c) the positive real numbers.

46. Determine the truth value of the statement $\exists x \forall y (x \leq y^2)$ if the domain for the variables consists of

- a) the positive real numbers.
- b) the integers.
- c) the nonzero real numbers.

47. Show that the two statements $\neg \exists x \forall y P(x, y)$ and $\forall x \exists y \neg P(x, y)$, where both quantifiers over the first variable in $P(x, y)$ have the same domain, and both quantifiers over the second variable in $P(x, y)$ have the same domain, are logically equivalent.

*48. Show that $\forall x P(x) \vee \forall x Q(x)$ and $\forall x \forall y (P(x) \vee Q(y))$, where all quantifiers have the same nonempty domain, are logically equivalent. (The new variable y is used to combine the quantifications correctly.)

*49. a) Show that $\forall x P(x) \wedge \exists x Q(x)$ is logically equivalent to $\forall x \exists y (P(x) \wedge Q(y))$, where all quantifiers have the same nonempty domain.

b) Show that $\forall x P(x) \vee \exists x Q(x)$ is equivalent to $\forall x \exists y (P(x) \vee Q(y))$, where all quantifiers have the same nonempty domain.

A statement is in **prenex normal form (PNF)** if and only if it is of the form

$$Q_1 x_1 Q_2 x_2 \cdots Q_k x_k P(x_1, x_2, \dots, x_k),$$

where each Q_i , $i = 1, 2, \dots, k$, is either the existential quantifier or the universal quantifier, and $P(x_1, \dots, x_k)$ is a predicate involving no quantifiers. For example, $\exists x \forall y (P(x, y) \wedge Q(y))$ is in prenex normal form, whereas $\exists x P(x) \vee \forall x Q(x)$ is not (because the quantifiers do not all occur first).

Every statement formed from propositional variables, predicates, **T**, and **F** using logical connectives and quantifiers is equivalent to a statement in prenex normal form. Exercise 51 asks for a proof of this fact.

*50. Put these statements in prenex normal form. [Hint: Use logical equivalence from Tables 6 and 7 in Section 1.3, Table 2 in Section 1.4, Example 19 in Section 1.4, Exercises 47 and 48 in Section 1.4, and Exercises 48 and 49.]

- a) $\exists x P(x) \vee \exists x Q(x) \vee A$, where A is a proposition not involving any quantifiers
- b) $\neg(\forall x P(x) \vee \forall x Q(x))$
- c) $\exists x P(x) \rightarrow \exists x Q(x)$

**51. Show how to transform an arbitrary statement to a statement in prenex normal form that is equivalent to the given statement. (Note: A formal solution of this exercise requires use of structural induction, covered in Section 5.3.)

*52. Express the quantification $\exists! x P(x)$, introduced in Section 1.4, using universal quantifications, existential quantifications, and logical operators.

The verification of universal modus tollens is left as Exercise 25. Exercises 26–29 develop additional combinations of rules of inference in propositional logic and quantified statements.

Exercises

1. Find the argument form for the following argument and determine whether it is valid. Can we conclude that the conclusion is true if the premises are true?

If Socrates is human, then Socrates is mortal.
Socrates is human.

∴ Socrates is mortal.

2. Find the argument form for the following argument and determine whether it is valid. Can we conclude that the conclusion is true if the premises are true?

If George does not have eight legs, then he is not a spider.
George is a spider.

∴ George has eight legs.

3. What rule of inference is used in each of these arguments?
- Alice is a mathematics major. Therefore, Alice is either a mathematics major or a computer science major.
 - Jerry is a mathematics major and a computer science major. Therefore, Jerry is a mathematics major.
 - If it is rainy, then the pool will be closed. It is rainy. Therefore, the pool is closed.
 - If it snows today, the university will close. The university is not closed today. Therefore, it did not snow today.
 - If I go swimming, then I will stay in the sun too long. If I stay in the sun too long, then I will sunburn. Therefore, if I go swimming, then I will sunburn.
4. What rule of inference is used in each of these arguments?
- Kangaroos live in Australia and are marsupials. Therefore, kangaroos are marsupials.
 - It is either hotter than 100 degrees today or the pollution is dangerous. It is less than 100 degrees outside today. Therefore, the pollution is dangerous.
 - Linda is an excellent swimmer. If Linda is an excellent swimmer, then she can work as a lifeguard. Therefore, Linda can work as a lifeguard.
 - Steve will work at a computer company this summer. Therefore, this summer Steve will work at a computer company or he will be a beach bum.
 - If I work all night on this homework, then I can answer all the exercises. If I answer all the exercises, I will understand the material. Therefore, if I work all night on this homework, then I will understand the material.

5. Use rules of inference to show that the hypotheses “Randy works hard,” “If Randy works hard, then he is a dull boy,” and “If Randy is a dull boy, then he will not get the job” imply the conclusion “Randy will not get the job.”

6. Use rules of inference to show that the hypotheses “If it does not rain or if it is not foggy, then the sailing race will be held and the lifesaving demonstration will go on,” “If the sailing race is held, then the trophy will be awarded,” and “The trophy was not awarded” imply the conclusion “It rained.”

7. What rules of inference are used in this famous argument? “All men are mortal. Socrates is a man. Therefore, Socrates is mortal.”

8. What rules of inference are used in this argument? “No man is an island. Manhattan is an island. Therefore, Manhattan is not a man.”

9. For each of these collections of premises, what relevant conclusion or conclusions can be drawn? Explain the rules of inference used to obtain each conclusion from the premises.

- “If I take the day off, it either rains or snows.” “I took Tuesday off or I took Thursday off.” “It was sunny on Tuesday.” “It did not snow on Thursday.”
- “If I eat spicy foods, then I have strange dreams.” “I have strange dreams if there is thunder while I sleep.” “I did not have strange dreams.”
- “I am either clever or lucky.” “I am not lucky.” “If I am lucky, then I will win the lottery.”
- “Every computer science major has a personal computer.” “Ralph does not have a personal computer.” “Ann has a personal computer.”
- “What is good for corporations is good for the United States.” “What is good for the United States is good for you.” “What is good for corporations is for you to buy lots of stuff.”
- “All rodents gnaw their food.” “Mice are rodents.” “Rabbits do not gnaw their food.” “Bats are not rodents.”

10. For each of these sets of premises, what relevant conclusion or conclusions can be drawn? Explain the rules of inference used to obtain each conclusion from the premises.

- “If I play hockey, then I am sore the next day.” “I use the whirlpool if I am sore.” “I did not use the whirlpool.”
- “If I work, it is either sunny or partly sunny.” “I worked last Monday or I worked last Friday.” “It was not sunny on Tuesday.” “It was not partly sunny on Friday.”
- “All insects have six legs.” “Dragonflies are insects.” “Spiders do not have six legs.” “Spiders eat dragonflies.”

- d) "Every student has an Internet account." "Homer does not have an Internet account." "Maggie has an Internet account."
- e) "All foods that are healthy to eat do not taste good." "Tofu is healthy to eat." "You only eat what tastes good." "You do not eat tofu." "Cheeseburgers are not healthy to eat."
- f) "I am either dreaming or hallucinating." "I am not dreaming." "If I am hallucinating, I see elephants running down the road."
11. Show that the argument form with premises $p_1, p_2, \dots, p_n$ and conclusion $q \rightarrow r$ is valid if the argument form with premises $p_1, p_2, \dots, p_n, q$, and conclusion r is valid.
12. Show that the argument form with premises $(p \wedge t) \rightarrow (r \vee s)$, $q \rightarrow (u \wedge t)$, $u \rightarrow p$, and $\neg s$ and conclusion $q \rightarrow r$ is valid by first using Exercise 11 and then using rules of inference from Table 1.
13. For each of these arguments, explain which rules of inference are used for each step.
- "Doug, a student in this class, knows how to write programs in JAVA. Everyone who knows how to write programs in JAVA can get a high-paying job. Therefore, someone in this class can get a high-paying job."
 - "Somebody in this class enjoys whale watching. Every person who enjoys whale watching cares about ocean pollution. Therefore, there is a person in this class who cares about ocean pollution."
 - "Each of the 93 students in this class owns a personal computer. Everyone who owns a personal computer can use a word processing program. Therefore, Zeke, a student in this class, can use a word processing program."
 - "Everyone in New Jersey lives within 50 miles of the ocean. Someone in New Jersey has never seen the ocean. Therefore, someone who lives within 50 miles of the ocean has never seen the ocean."
14. For each of these arguments, explain which rules of inference are used for each step.
- "Linda, a student in this class, owns a red convertible. Everyone who owns a red convertible has gotten at least one speeding ticket. Therefore, someone in this class has gotten a speeding ticket."
 - "Each of five roommates, Melissa, Aaron, Ralph, Veneesha, and Keeshawn, has taken a course in discrete mathematics. Every student who has taken a course in discrete mathematics can take a course in algorithms. Therefore, all five roommates can take a course in algorithms next year."
 - "All movies produced by John Sayles are wonderful. John Sayles produced a movie about coal miners. Therefore, there is a wonderful movie about coal miners."
 - "There is someone in this class who has been to France. Everyone who goes to France visits the Louvre. Therefore, someone in this class has visited the Louvre."
15. For each of these arguments determine whether the argument is correct or incorrect and explain why.
- All students in this class understand logic. Xavier is a student in this class. Therefore, Xavier understands logic.
 - Every computer science major takes discrete mathematics. Natasha is taking discrete mathematics. Therefore, Natasha is a computer science major.
 - All parrots like fruit. My pet bird is not a parrot. Therefore, my pet bird does not like fruit.
 - Everyone who eats granola every day is healthy. Linda is not healthy. Therefore, Linda does not eat granola every day.
16. For each of these arguments determine whether the argument is correct or incorrect and explain why.
- Everyone enrolled in the university has lived in a dormitory. Mia has never lived in a dormitory. Therefore, Mia is not enrolled in the university.
 - A convertible car is fun to drive. Isaac's car is not a convertible. Therefore, Isaac's car is not fun to drive.
 - Quincy likes all action movies. Quincy likes the movie *Eight Men Out*. Therefore, *Eight Men Out* is an action movie.
 - All lobstermen set at least a dozen traps. Hamilton is a lobsterman. Therefore, Hamilton sets at least a dozen traps.
17. What is wrong with this argument? Let $H(x)$ be " x is happy." Given the premise $\exists x H(x)$, we conclude that $H(\text{Lola})$. Therefore, Lola is happy.
18. What is wrong with this argument? Let $S(x, y)$ be " x is shorter than y ." Given the premise $\exists s S(s, \text{Max})$, it follows that $S(\text{Max}, \text{Max})$. Then by existential generalization it follows that $\exists x S(x, x)$, so that someone is shorter than himself.
19. Determine whether each of these arguments is valid. If an argument is correct, what rule of inference is being used? If it is not, what logical error occurs?
- If n is a real number such that $n > 1$, then $n^2 > 1$. Suppose that $n^2 > 1$. Then $n > 1$.
 - If n is a real number with $n > 3$, then $n^2 > 9$. Suppose that $n^2 \leq 9$. Then $n \leq 3$.
 - If n is a real number with $n > 2$, then $n^2 > 4$. Suppose that $n \leq 2$. Then $n^2 \leq 4$.
20. Determine whether these are valid arguments.
- If x is a positive real number, then x^2 is a positive real number. Therefore, if a^2 is positive, where a is a real number, then a is a positive real number.
 - If $x^2 \neq 0$, where x is a real number, then $x \neq 0$. Let a be a real number with $a^2 \neq 0$; then $a \neq 0$.
21. Which rules of inference are used to establish the conclusion of Lewis Carroll's argument described in Example 26 of Section 1.4?
22. Which rules of inference are used to establish the conclusion of Lewis Carroll's argument described in Example 27 of Section 1.4?

23. Identify the error or errors in this argument that supposedly shows that if $\exists xP(x) \wedge \exists xQ(x)$ is true then $\exists x(P(x) \wedge Q(x))$ is true.

1. $\exists xP(x) \vee \exists xQ(x)$ Premise
2. $\exists xP(x)$ Simplification from (1)
3. $P(c)$ Existential instantiation from (2)
4. $\exists xQ(x)$ Simplification from (1)
5. $Q(c)$ Existential instantiation from (4)
6. $P(c) \wedge Q(c)$ Conjunction from (3) and (5)
7. $\exists x(P(x) \wedge Q(x))$ Existential generalization

24. Identify the error or errors in this argument that supposedly shows that if $\forall x(P(x) \vee Q(x))$ is true then $\forall xP(x) \vee \forall xQ(x)$ is true.

1. $\forall x(P(x) \vee Q(x))$ Premise
2. $P(c) \vee Q(c)$ Universal instantiation from (1)
3. $P(c)$ Simplification from (2)
4. $\forall xP(x)$ Universal generalization from (3)
5. $Q(c)$ Simplification from (2)
6. $\forall xQ(x)$ Universal generalization from (5)
7. $\forall x(P(x) \vee \forall xQ(x))$ Conjunction from (4) and (6)

25. Justify the rule of universal modus tollens by showing that the premises $\forall x(P(x) \rightarrow Q(x))$ and $\neg Q(a)$ for a particular element a in the domain, imply $\neg P(a)$.

26. Justify the rule of **universal transitivity**, which states that if $\forall x(P(x) \rightarrow Q(x))$ and $\forall x(Q(x) \rightarrow R(x))$ are true, then $\forall x(P(x) \rightarrow R(x))$ is true, where the domains of all quantifiers are the same.

27. Use rules of inference to show that if $\forall x(P(x) \rightarrow (Q(x) \wedge S(x)))$ and $\forall x(P(x) \wedge R(x))$ are true, then $\forall x(R(x) \wedge S(x))$ is true.

28. Use rules of inference to show that if $\forall x(P(x) \vee Q(x))$ and $\forall x((\neg P(x) \wedge Q(x)) \rightarrow R(x))$ are true, then $\forall x(\neg R(x) \rightarrow P(x))$ is also true, where the domains of all quantifiers are the same.

29. Use rules of inference to show that if $\forall x(P(x) \vee Q(x))$, $\forall x(\neg Q(x) \vee S(x))$, $\forall x(R(x) \rightarrow \neg S(x))$, and $\exists x\neg P(x)$ are true, then $\exists x\neg R(x)$ is true.

30. Use resolution to show the hypotheses “Allen is a bad boy or Hillary is a good girl” and “Allen is a good boy or

David is happy” imply the conclusion “Hillary is a good girl or David is happy.”

31. Use resolution to show that the hypotheses “It is not raining or Yvette has her umbrella,” “Yvette does not have her umbrella or she does not get wet,” and “It is raining or Yvette does not get wet” imply that “Yvette does not get wet.”

32. Show that the equivalence $p \wedge \neg p \equiv \mathbf{F}$ can be derived using resolution together with the fact that a conditional statement with a false hypothesis is true. [Hint: Let $q = r = \mathbf{F}$ in resolution.]

33. Use resolution to show that the compound proposition $(p \vee q) \wedge (\neg p \vee q) \wedge (p \vee \neg q) \wedge (\neg p \vee \neg q)$ is not satisfiable.

- *34. The Logic Problem, taken from *WFF'N PROOF, The Game of Logic*, has these two assumptions:

1. “Logic is difficult or not many students like logic.”
2. “If mathematics is easy, then logic is not difficult.”

By translating these assumptions into statements involving propositional variables and logical connectives, determine whether each of the following are valid conclusions of these assumptions:

- a) That mathematics is not easy, if many students like logic.
- b) That not many students like logic, if mathematics is not easy.
- c) That mathematics is not easy or logic is difficult.
- d) That logic is not difficult or mathematics is not easy.
- e) That if not many students like logic, then either mathematics is not easy or logic is not difficult.

- *35. Determine whether this argument, taken from Kalish and Montague [KaMo64], is valid.

If Superman were able and willing to prevent evil, he would do so. If Superman were unable to prevent evil, he would be impotent; if he were unwilling to prevent evil, he would be malevolent. Superman does not prevent evil. If Superman exists, he is neither impotent nor malevolent. Therefore, Superman does not exist.

1.7 Introduction to Proofs

1.7.1 Introduction

In this section we introduce the notion of a proof and describe methods for constructing proofs. A proof is a valid argument that establishes the truth of a mathematical statement. A proof can use the hypotheses of the theorem, if any, axioms assumed to be true, and previously proven theorems. Using these ingredients and rules of inference, the final step of the proof establishes the truth of the statement being proved.

In our discussion we move from formal proofs of theorems toward more informal proofs. The arguments we introduced in Section 1.6 to show that statements involving propositions and quantified statements are true were formal proofs, where all steps were supplied, and the rules for each step in the argument were given. However, formal proofs of useful theorems can

Making mistakes in proofs is part of the learning process. When you make a mistake that someone else finds, you should carefully analyze where you went wrong and make sure that you do not make the same mistake again. Even professional mathematicians make mistakes in proofs. More than a few incorrect proofs of important results have fooled people for many years before subtle errors in them were found.

1.7.9 Just a Beginning

We have now developed a basic arsenal of proof methods. In the next section we will introduce other important proof methods. We will also introduce several important proof techniques in Chapter 5, including mathematical induction, which can be used to prove results that hold for all positive integers. In Chapter 6 we will introduce the notion of combinatorial proofs.

In this section we introduced several methods for proving theorems of the form $\forall x(P(x) \rightarrow Q(x))$, including direct proofs and proofs by contraposition. There are many theorems of this type whose proofs are easy to construct by directly working through the hypotheses and definitions of the terms of the theorem. However, it is often difficult to prove a theorem without resorting to a clever use of a proof by contraposition or a proof by contradiction, or some other proof technique. In Section 1.8 we will address proof strategy. We will describe various approaches that can be used to find proofs when straightforward approaches do not work. Constructing proofs is an art that can be learned only through experience, including writing proofs, having your proofs critiqued, and reading and analyzing other proofs.

Exercises

1. Use a direct proof to show that the sum of two odd integers is even.
2. Use a direct proof to show that the sum of two even integers is even.
3. Show that the square of an even number is an even number using a direct proof.
4. Show that the additive inverse, or negative, of an even number is an even number using a direct proof.
5. Prove that if $m + n$ and $n + p$ are even integers, where m , n , and p are integers, then $m + p$ is even. What kind of proof did you use?
6. Use a direct proof to show that the product of two odd numbers is odd.
7. Use a direct proof to show that every odd integer is the difference of two squares. [Hint: Find the difference of the squares of $k + 1$ and k where k is a positive integer.]
8. Prove that if n is a perfect square, then $n + 2$ is not a perfect square.
9. Use a proof by contradiction to prove that the sum of an irrational number and a rational number is irrational.
10. Use a direct proof to show that the product of two rational numbers is rational.
11. Prove or disprove that the product of two irrational numbers is irrational.
12. Prove or disprove that the product of a nonzero rational number and an irrational number is irrational.
13. Prove that if x is irrational, then $1/x$ is irrational.
14. Prove that if x is rational and $x \neq 0$, then $1/x$ is rational.
15. Prove that if x is an irrational number and $x > 0$, then $\sqrt{x}$ is also irrational.
16. Prove that if x , y , and z are integers and $x + y + z$ is odd, then at least one of x , y , and z is odd.
17. Use a proof by contraposition to show that if $x + y \geq 2$, where x and y are real numbers, then $x \geq 1$ or $y \geq 1$.
18. Prove that if m and n are integers and mn is even, then m is even or n is even.
19. Show that if n is an integer and $n^3 + 5$ is odd, then n is even using
 - a) a proof by contraposition.
 - b) a proof by contradiction.
20. Prove that if n is an integer and $3n + 2$ is even, then n is even using
 - a) a proof by contraposition.
 - b) a proof by contradiction.
21. Prove the proposition $P(0)$, where $P(n)$ is the proposition "If n is a positive integer greater than 1, then $n^2 > n$." What kind of proof did you use?
22. Prove the proposition $P(1)$, where $P(n)$ is the proposition "If n is a positive integer, then $n^2 \geq n$." What kind of proof did you use?
23. Let $P(n)$ be the proposition "If a and b are positive real numbers, then $(a + b)^n \geq a^n + b^n$." Prove that $P(1)$ is true. What kind of proof did you use?
24. Show that if you pick three socks from a drawer containing just blue socks and black socks, you must get either a pair of blue socks or a pair of black socks.

25. Show that at least ten of any 64 days chosen must fall on the same day of the week.
26. Show that at least three of any 25 days chosen must fall in the same month of the year.
27. Use a proof by contradiction to show that there is no rational number r for which $r^3 + r + 1 = 0$. [Hint: Assume that $r = a/b$ is a root, where a and b are integers and a/b is in lowest terms. Obtain an equation involving integers by multiplying by b^3 . Then look at whether a and b are each odd or even.]
28. Prove that if n is a positive integer, then n is even if and only if $7n + 4$ is even.
29. Prove that if n is a positive integer, then n is odd if and only if $5n + 6$ is odd.
30. Prove that $m^2 = n^2$ if and only if $m = n$ or $m = -n$.
31. Prove or disprove that if m and n are integers such that $mn = 1$, then either $m = 1$ and $n = 1$, or else $m = -1$ and $n = -1$.
32. Show that these three statements are equivalent, where a and b are real numbers: (i) a is less than b , (ii) the average of a and b is greater than a , and (iii) the average of a and b is less than b .
33. Show that these statements about the integer x are equivalent: (i) $3x + 2$ is even, (ii) $x + 5$ is odd, (iii) x^2 is even.
34. Show that these statements about the real number x are equivalent: (i) x is rational, (ii) $x/2$ is rational, (iii) $3x - 1$ is rational.
35. Show that these statements about the real number x are equivalent: (i) x is irrational, (ii) $3x + 2$ is irrational, (iii) $x/2$ is irrational.
36. Is this reasoning for finding the solutions of the equation $\sqrt{2x^2 - 1} = x$ correct? (1) $\sqrt{2x^2 - 1} = x$ is given; (2) $2x^2 - 1 = x^2$, obtained by squaring both sides of (1); (3) $x^2 - 1 = 0$, obtained by subtracting x^2 from both sides of (2); (4) $(x - 1)(x + 1) = 0$, obtained by factoring the left-hand side of $x^2 - 1$; (5) $x = 1$ or $x = -1$, which follows because $ab = 0$ implies that $a = 0$ or $b = 0$.
37. Are these steps for finding the solutions of $\sqrt{x + 3} = 3 - x$ correct? (1) $\sqrt{x + 3} = 3 - x$ is given; (2) $x + 3 = x^2 - 6x + 9$, obtained by squaring both sides of (1); (3) $0 = x^2 - 7x + 6$, obtained by subtracting $x + 3$ from both sides of (2); (4) $0 = (x - 1)(x - 6)$, obtained by factoring the right-hand side of (3); (5) $x = 1$ or $x = 6$, which follows from (4) because $ab = 0$ implies that $a = 0$ or $b = 0$.
38. Show that the propositions p_1, p_2, p_3 , and p_4 can be shown to be equivalent by showing that $p_1 \leftrightarrow p_4, p_2 \leftrightarrow p_3$, and $p_1 \leftrightarrow p_3$.
39. Show that the propositions p_1, p_2, p_3, p_4 , and p_5 can be shown to be equivalent by proving that the conditional statements $p_1 \rightarrow p_4, p_3 \rightarrow p_1, p_4 \rightarrow p_2, p_2 \rightarrow p_5$, and $p_5 \rightarrow p_3$ are true.
40. Find a counterexample to the statement that every positive integer can be written as the sum of the squares of three integers.
41. Prove that at least one of the real numbers $a_1, a_2, \dots, a_n$ is greater than or equal to the average of these numbers. What kind of proof did you use?
42. Use Exercise 41 to show that if the first 10 positive integers are placed around a circle, in any order, there exist three integers in consecutive locations around the circle that have a sum greater than or equal to 17.
43. Prove that if n is an integer, these four statements are equivalent: (i) n is even, (ii) $n + 1$ is odd, (iii) $3n + 1$ is odd, (iv) $3n$ is even.
44. Prove that these four statements about the integer n are equivalent: (i) n^2 is odd, (ii) $1 - n$ is even, (iii) n^3 is odd, (iv) $n^2 + 1$ is even.

1.8 Proof Methods and Strategy

1.8.1 Introduction

Assessment

In Section 1.7 we introduced many methods of proof and illustrated how each method can be used. In this section we continue this effort. We will introduce several other commonly used proof methods, including the method of proving a theorem by considering different cases separately. We will also discuss proofs where we prove the existence of objects with desired properties.

In Section 1.7 we briefly discussed the strategy behind constructing proofs. This strategy includes selecting a proof method and then successfully constructing an argument step by step, based on this method. In this section, after we have developed a versatile arsenal of proof methods, we will study some aspects of the art and science of proofs. We will provide advice on how to find a proof of a theorem. We will describe some tricks of the trade, including how proofs can be found by working backward and by adapting existing proofs.

4.3, where these open questions are discussed.) You will encounter many other open questions as you read this book. The study of such problems has played and continues to play an important role in the development of many parts of discrete mathematics.

Build up your arsenal of proof methods as you work through this book.

1.8.10 Additional Proof Methods

In this chapter we introduced the basic methods used in proofs. We also described how to leverage these methods to prove a variety of results. We will use these proof methods in all subsequent chapters. In particular, we will use them in Chapters 2, 3, and 4 to prove results about sets, functions, algorithms, and number theory and in Chapters 9, 10, and 11 to prove results in graph theory. Among the theorems we will prove is the famous halting theorem, which states that there is a problem that cannot be solved using any procedure. However, there are many important proof methods besides those we have covered. We will introduce some of these methods later in this book. In particular, in Section 5.1 we will discuss mathematical induction, which is an extremely useful method for proving statements of the form $\forall nP(n)$, where the domain consists of all positive integers. In Section 5.3 we will introduce structural induction, which can be used to prove results about recursively defined sets. We will use the Cantor diagonalization method, which can be used to prove results about the size of infinite sets, in Section 2.5. In Chapter 6 we will introduce the notion of combinatorial proofs, which can be used to prove results by counting arguments. The reader should note that entire books have been devoted to the activities discussed in this section, including many excellent works by George Pólya ([Po61], [Po71], [Po90]).

Finally, note that we have not given a procedure that can be used for proving theorems in mathematics. It is a deep theorem of mathematical logic that there is no such procedure.

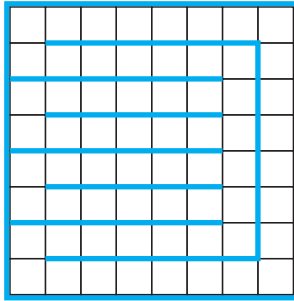
Exercises

1. Prove that $n^2 + 1 \geq 2^n$ when n is a positive integer with $1 \leq n \leq 4$.
2. Use a proof by cases to show that 10 is not the square of a positive integer. [Hint: Consider two cases: (i) $1 \leq x \leq 3$, (ii) $x \geq 4$.]
3. Use a proof by cases to show that 100 is not the cube of a positive integer. [Hint: Consider two cases: (i) $1 \leq x \leq 4$, (ii) $x \geq 5$.]
4. Prove that there are no positive perfect cubes less than 1000 that are the sum of the cubes of two positive integers.
5. Prove that if x and y are real numbers, then $\max(x, y) + \min(x, y) = x + y$. [Hint: Use a proof by cases, with the two cases corresponding to $x \geq y$ and $x < y$, respectively.]
6. Use a proof by cases to show that $\min(a, \min(b, c)) = \min(\min(a, b), c)$ whenever a , b , and c are real numbers.
7. Prove using the notion of without loss of generality that $\min(x, y) = (x + y - |x - y|)/2$ and $\max(x, y) = (x + y + |x - y|)/2$ whenever x and y are real numbers.
8. Prove using the notion of without loss of generality that $5x + 5y$ is an odd integer when x and y are integers of opposite parity.
9. Prove the **triangle inequality**, which states that if x and y are real numbers, then $|x| + |y| \geq |x + y|$ (where $|x|$ represents the absolute value of x , which equals x if $x \geq 0$ and equals $-x$ if $x < 0$).
10. Prove that there is a positive integer that equals the sum of the positive integers not exceeding it. Is your proof constructive or nonconstructive?
11. Prove that there are 100 consecutive positive integers that are not perfect squares. Is your proof constructive or nonconstructive?
12. Prove that either $2 \cdot 10^{500} + 15$ or $2 \cdot 10^{500} + 16$ is not a perfect square. Is your proof constructive or nonconstructive?
13. Prove that there exists a pair of consecutive integers such that one of these integers is a perfect square and the other is a perfect cube.
14. Show that the product of two of the numbers $65^{1000} - 8^{2001} + 3^{177}$, $79^{1212} - 9^{2399} + 2^{2001}$, and $24^{4493} - 5^{8192} + 7^{1777}$ is nonnegative. Is your proof constructive or nonconstructive? [Hint: Do not try to evaluate these numbers!]
15. Prove or disprove that there is a rational number x and an irrational number y such that x^y is irrational.
16. Prove or disprove that if a and b are rational numbers, then a^b is also rational.

17. Show that each of these statements can be used to express the fact that there is a unique element x such that $P(x)$ is true. [Note that we can also write this statement as $\exists!xP(x)$.]
 - a) $\exists x\forall y(P(y) \leftrightarrow x = y)$
 - b) $\exists xP(x) \wedge \forall x\forall y(P(x) \wedge P(y) \rightarrow x = y)$
 - c) $\exists x(P(x) \wedge \forall y(P(y) \rightarrow x = y))$
18. Show that if a , b , and c are real numbers and $a \neq 0$, then there is a unique solution of the equation $ax + b = c$.
19. Suppose that a and b are odd integers with $a \neq b$. Show there is a unique integer c such that $|a - c| = |b - c|$.
20. Show that if r is an irrational number, there is a unique integer n such that the distance between r and n is less than $1/2$.
21. Show that if n is an odd integer, then there is a unique integer k such that n is the sum of $k - 2$ and $k + 3$.
22. Prove that given a real number x there exist unique numbers n and ϵ such that $x = n + \epsilon$, n is an integer, and $0 \leq \epsilon < 1$.
23. Prove that given a real number x there exist unique numbers n and ϵ such that $x = n - \epsilon$, n is an integer, and $0 \leq \epsilon < 1$.
24. Use forward reasoning to show that if x is a nonzero real number, then $x^2 + 1/x^2 \geq 2$. [Hint: Start with the inequality $(x - 1/x)^2 \geq 0$, which holds for all nonzero real numbers x .]
25. The **harmonic mean** of two real numbers x and y equals $2xy/(x + y)$. By computing the harmonic and geometric means of different pairs of positive real numbers, formulate a conjecture about their relative sizes and prove your conjecture.
26. The **quadratic mean** of two real numbers x and y equals $\sqrt{(x^2 + y^2)/2}$. By computing the arithmetic and quadratic means of different pairs of positive real numbers, formulate a conjecture about their relative sizes and prove your conjecture.
- *27. Write the numbers $1, 2, \dots, 2n$ on a blackboard, where n is an odd integer. Pick any two of the numbers, j and k , write $|j - k|$ on the board and erase j and k . Continue this process until only one integer is written on the board. Prove that this integer must be odd.
- *28. Suppose that five ones and four zeros are arranged around a circle. Between any two equal bits you insert a 0 and between any two unequal bits you insert a 1 to produce nine new bits. Then you erase the nine original bits. Show that when you iterate this procedure, you can never get nine zeros. [Hint: Work backward, assuming that you did end up with nine zeros.]
29. Formulate a conjecture about the decimal digits that appear as the final decimal digit of the fourth power of an integer. Prove your conjecture using a proof by cases.
30. Formulate a conjecture about the final two decimal digits of the square of an integer. Prove your conjecture using a proof by cases.
31. Prove that there is no positive integer n such that $n^2 + n^3 = 100$.
32. Prove that there are no solutions in integers x and y to the equation $2x^2 + 5y^2 = 14$.
33. Prove that there are no solutions in positive integers x and y to the equation $x^4 + y^4 = 625$.
34. Prove that there are infinitely many solutions in positive integers x , y , and z to the equation $x^2 + y^2 = z^2$. [Hint: Let $x = m^2 - n^2$, $y = 2mn$, and $z = m^2 + n^2$, where m and n are integers.]
35. Adapt the proof in Example 4 in Section 1.7 to prove that if $n = abc$, where a , b , and c are positive integers, then $a \leq \sqrt[3]{n}$, $b \leq \sqrt[3]{n}$, or $c \leq \sqrt[3]{n}$.
36. Prove that $\sqrt[3]{2}$ is irrational.
37. Prove that between every two rational numbers there is an irrational number.
38. Prove that between every rational number and every irrational number there is an irrational number.
- *39. Let $S = x_1y_1 + x_2y_2 + \dots + x_ny_n$, where $x_1, x_2, \dots, x_n$ and $y_1, y_2, \dots, y_n$ are orderings of two different sequences of positive real numbers, each containing n elements.
 - a) Show that S takes its maximum value over all orderings of the two sequences when both sequences are sorted (so that the elements in each sequence are in nondecreasing order).
 - b) Show that S takes its minimum value over all orderings of the two sequences when one sequence is sorted into nondecreasing order and the other is sorted into nonincreasing order.
40. Prove or disprove that if you have an 8-gallon jug of water and two empty jugs with capacities of 5 gallons and 3 gallons, respectively, then you can measure 4 gallons by successively pouring some of or all of the water in a jug into another jug.
41. Verify the $3x + 1$ conjecture for these integers.
 - a) 6 b) 7 c) 17 d) 21
42. Verify the $3x + 1$ conjecture for these integers.
 - a) 16 b) 11 c) 35 d) 113
43. Prove or disprove that you can use dominoes to tile the standard checkerboard with two adjacent corners removed (that is, corners that are not opposite).
44. Prove or disprove that you can use dominoes to tile a standard checkerboard with all four corners removed.
45. Prove that you can use dominoes to tile a rectangular checkerboard with an even number of squares.
46. Prove or disprove that you can use dominoes to tile a 5×5 checkerboard with three corners removed.
47. Use a proof by exhaustion to show that a tiling using dominoes of a 4×4 checkerboard with opposite corners removed does not exist. [Hint: First show that you can assume that the squares in the upper left and lower right corners are removed. Number the squares of the original checkerboard from 1 to 16, starting in the first row, moving right in this row, then starting in the leftmost square in the second row and moving right, and so on. Remove squares 1 and 16. To begin the proof, note that square 2 is covered either by a domino laid horizontally, which

covers squares 2 and 3, or vertically, which covers squares 2 and 6. Consider each of these cases separately, and work through all the subcases that arise.]

- *48. Prove that when a white square and a black square are removed from an 8×8 checkerboard (colored as in the text) you can tile the remaining squares of the checkerboard using dominoes. [Hint: Show that when one black and one white square are removed, each part of the partition of the remaining cells formed by inserting the barriers shown in the figure can be covered by dominoes.]



49. Show that by removing two white squares and two black squares from an 8×8 checkerboard (colored as in the text) you can make it impossible to tile the remaining squares using dominoes.

- *50. Find all squares, if they exist, on an 8×8 checkerboard such that the board obtained by removing one of these squares can be tiled using straight triominoes. [Hint: First use arguments based on coloring and rotations to eliminate as many squares as possible from consideration.]

- *51. a) Draw each of the five different tetrominoes, where a tetromino is a polyomino consisting of four squares.

- b) For each of the five different tetrominoes, prove or disprove that you can tile a standard checkerboard using these tetrominoes.

- *52. Prove or disprove that you can tile a 10×10 checkerboard using straight tetrominoes.

Key Terms and Results

TERMS

proposition: a statement that is true or false

propositional variable: a variable that represents a proposition

truth value: true or false

$\neg p$ (**negation of p**): the proposition with truth value opposite to the truth value of p

logical operators: operators used to combine propositions

compound proposition: a proposition constructed by combining propositions using logical operators

truth table: a table displaying all possible truth values of propositions

$p \vee q$ (**disjunction of p and q**): the proposition “ p or q ,” which is true if and only if at least one of p and q is true

$p \wedge q$ (**conjunction of p and q**): the proposition “ p and q ,” which is true if and only if both p and q are true

$p \oplus q$ (**exclusive or of p and q**): the proposition “ p XOR q ,” which is true when exactly one of p and q is true

$p \rightarrow q$ (**p implies q**): the proposition “if p , then q ,” which is false if and only if p is true and q is false

converse of $p \rightarrow q$: the conditional statement $q \rightarrow p$

contrapositive of $p \rightarrow q$: the conditional statement $\neg q \rightarrow \neg p$

inverse of $p \rightarrow q$: the conditional statement $\neg p \rightarrow \neg q$

$p \leftrightarrow q$ (**biconditional**): the proposition “ p if and only if q ,” which is true if and only if p and q have the same truth value

bit: either a 0 or a 1

Boolean variable: a variable that has a value of 0 or 1

bit operation: an operation on a bit or bits

bit string: a list of bits

bitwise operations: operations on bit strings that operate on each bit in one string and the corresponding bit in the other string

logic gate: a logic element that performs a logical operation on one or more bits to produce an output bit

logic circuit: a switching circuit made up of logic gates that produces one or more output bits

tautology: a compound proposition that is always true

contradiction: a compound proposition that is always false

contingency: a compound proposition that is sometimes true and sometimes false

consistent compound propositions: compound propositions for which there is an assignment of truth values to the variables that makes all these propositions true

satisfiable compound proposition: a compound proposition for which there is an assignment of truth values to its variables that makes it true

logically equivalent compound propositions: compound propositions that always have the same truth values

predicate: part of a sentence that attributes a property to the subject

propositional function: a statement containing one or more variables that becomes a proposition when each of its variables is assigned a value or is bound by a quantifier

domain (or universe) of discourse: the values a variable in a propositional function may take

$\exists x P(x)$ (existential quantification of $P(x)$): the proposition that is true if and only if there exists an x in the domain such that $P(x)$ is true

$\forall x P(x)$ (universal quantification of $P(x)$): the proposition that is true if and only if $P(x)$ is true for every x in the domain

logically equivalent expressions: expressions that have the same truth value no matter which propositional functions and domains are used

free variable: a variable not bound in a propositional function

bound variable: a variable that is quantified

scope of a quantifier: portion of a statement where the quantifier binds its variable

argument: a sequence of statements

argument form: a sequence of compound propositions involving propositional variables

premise: a statement, in an argument, or argument form, other than the final one

conclusion: the final statement in an argument or argument form

valid argument form: a sequence of compound propositions involving propositional variables where the truth of all the premises implies the truth of the conclusion

valid argument: an argument with a valid argument form

rule of inference: a valid argument form that can be used in the demonstration that arguments are valid

fallacy: an invalid argument form often used incorrectly as a rule of inference (or sometimes, more generally, an incorrect argument)

circular reasoning or begging the question: reasoning where one or more steps are based on the truth of the statement being proved

theorem: a mathematical assertion that can be shown to be true

conjecture: a mathematical assertion proposed to be true, but that has not been proved

proof: a demonstration that a theorem is true

axiom: a statement that is assumed to be true and that can be used as a basis for proving theorems

lemma: a theorem used to prove other theorems

corollary: a proposition that can be proved as a consequence of a theorem that has just been proved

vacuous proof: a proof that $p \rightarrow q$ is true based on the fact that p is false

trivial proof: a proof that $p \rightarrow q$ is true based on the fact that q is true

direct proof: a proof that $p \rightarrow q$ is true that proceeds by showing that q must be true when p is true

proof by contraposition: a proof that $p \rightarrow q$ is true that proceeds by showing that p must be false when q is false

proof by contradiction: a proof that p is true based on the truth of the conditional statement $\neg p \rightarrow q$, where q is a contradiction

exhaustive proof: a proof that establishes a result by checking a list of all possible cases

proof by cases: a proof broken into separate cases, where these cases cover all possibilities

without loss of generality: an assumption in a proof that makes it possible to prove a theorem by reducing the number of cases to consider in the proof

counterexample: an element x such that $P(x)$ is false

constructive existence proof: a proof that an element with a specified property exists that explicitly finds such an element

nonconstructive existence proof: a proof that an element with a specified property exists that does not explicitly find such an element

rational number: a number that can be expressed as the ratio of two integers p and q such that $q \neq 0$

uniqueness proof: a proof that there is exactly one element satisfying a specified property

RESULTS

The logical equivalences given in Tables 6, 7, and 8 in Section 1.3.

De Morgan's laws for quantifiers.

Rules of inference for propositional calculus.

Rules of inference for quantified statements.

Review Questions

- Define the negation of a proposition.
 - What is the negation of "This is a boring course"?
- Define (using truth tables) the disjunction, conjunction, exclusive or, conditional, and biconditional of the propositions p and q .
 - What are the disjunction, conjunction, exclusive or, conditional, and biconditional of the propositions "I'll go to the movies tonight" and "I'll finish my discrete mathematics homework"?
- Describe at least five different ways to write the conditional statement $p \rightarrow q$ in English.
 - Define the converse and contrapositive of a conditional statement.
 - State the converse and the contrapositive of the conditional statement "If it is sunny tomorrow, then I will go for a walk in the woods."
- What does it mean for two propositions to be logically equivalent?
 - Describe the different ways to show that two compound propositions are logically equivalent.
 - Show in at least two different ways that the compound propositions $\neg p \vee (r \rightarrow \neg q)$ and $\neg p \vee \neg q \vee \neg r$ are equivalent.

5. (Depends on the Exercise Set in Section 1.3)
- Given a truth table, explain how to use disjunctive normal form to construct a compound proposition with this truth table.
 - Explain why part (a) shows that the operators $\wedge$, $\vee$, and $\neg$ are functionally complete.
 - Is there an operator such that the set containing just this operator is functionally complete?
6. What are the universal and existential quantifications of a predicate $P(x)$? What are their negations?
7. a) What is the difference between the quantification $\exists x \forall y P(x, y)$ and $\forall y \exists x P(x, y)$, where $P(x, y)$ is a predicate?
- b) Give an example of a predicate $P(x, y)$ such that $\exists x \forall y P(x, y)$ and $\forall y \exists x P(x, y)$ have different truth values.
8. Describe what is meant by a valid argument in propositional logic and show that the argument “If the earth is flat, then you can sail off the edge of the earth,” “You cannot sail off the edge of the earth,” therefore, “The earth is not flat” is a valid argument.
9. Use rules of inference to show that if the premises “All zebras have stripes” and “Mark is a zebra” are true, then the conclusion “Mark has stripes” is true.
10. a) Describe what is meant by a direct proof, a proof by contraposition, and a proof by contradiction of a conditional statement $p \rightarrow q$.
- b) Give a direct proof, a proof by contraposition, and a proof by contradiction of the statement: “If n is even, then $n + 4$ is even.”
11. a) Describe a way to prove the biconditional $p \leftrightarrow q$.
- b) Prove the statement: “The integer $3n + 2$ is odd if and only if the integer $9n + 5$ is even, where n is an integer.”
12. To prove that the statements p_1, p_2, p_3 , and p_4 are equivalent, is it sufficient to show that the conditional statements $p_4 \rightarrow p_2, p_3 \rightarrow p_1$, and $p_1 \rightarrow p_2$ are valid? If not, provide another collection of conditional statements that can be used to show that the four statements are equivalent.
13. a) Suppose that a statement of the form $\forall x P(x)$ is false. How can this be proved?
- b) Show that the statement “For every positive integer n , $n^2 \geq 2n$ ” is false.
14. What is the difference between a constructive and non-constructive existence proof? Give an example of each.
15. What are the elements of a proof that there is a unique element x such that $P(x)$, where $P(x)$ is a propositional function?
16. Explain how a proof by cases can be used to prove a result about absolute values, such as the fact that $|xy| = |x||y|$ for all real numbers x and y .

Supplementary Exercises

- Let p be the proposition “I will do every exercise in this book” and q be the proposition “I will get an A in this course.” Express each of these as a combination of p and q .
 - I will get an A in this course only if I do every exercise in this book.
 - I will get an A in this course and I will do every exercise in this book.
 - Either I will not get an A in this course or I will not do every exercise in this book.
 - For me to get an A in this course it is necessary and sufficient that I do every exercise in this book.
 - Find the truth table of the compound proposition $(p \vee q) \rightarrow (p \wedge \neg r)$.
 - Show that these compound propositions are tautologies.
 - $(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$
 - $((p \vee q) \wedge \neg p) \rightarrow q$
 - Give the converse, the contrapositive, and the inverse of these conditional statements.
 - If it rains today, then I will drive to work.
 - If $|x| = x$, then $x \geq 0$.
 - If n is greater than 3, then n^2 is greater than 9.
 - Given a conditional statement $p \rightarrow q$, find the converse of its inverse, the converse of its converse, and the converse of its contrapositive.
 - Given a conditional statement $p \rightarrow q$, find the inverse of its inverse, the inverse of its converse, and the inverse of its contrapositive.
 - Find a compound proposition involving the propositional variables p, q, r , and s that is true when exactly three of these propositional variables are true and is false otherwise.
 - Show that these statements are inconsistent: “If Sergei takes the job offer, then he will get a signing bonus.” “If Sergei takes the job offer, then he will receive a higher salary.” “If Sergei gets a signing bonus, then he will not receive a higher salary.” “Sergei takes the job offer.”
 - Show that these statements are inconsistent: “If Miranda does not take a course in discrete mathematics, then she will not graduate.” “If Miranda does not graduate, then she is not qualified for the job.” “If Miranda reads this book, then she is qualified for the job.” “Miranda does not take a course in discrete mathematics but she reads this book.”
- Teachers in the Middle Ages supposedly tested the realtime propositional logic ability of a student via a technique known as an **obligato game**. In an obligato game, a number of rounds is set and in each round the teacher gives the student successive assertions that the student must either accept or reject as they are given. When the student accepts an assertion, it is

added as a commitment; when the student rejects an assertion its negation is added as a commitment. The student passes the test if the consistency of all commitments is maintained throughout the test.

10. Suppose that in a three-round obligato game, the teacher first gives the student the proposition $p \rightarrow q$, then the proposition $\neg(p \vee r) \vee q$, and finally, the proposition q . For which of the eight possible sequences of three answers will the student pass the test?
11. Suppose that in a four-round obligato game, the teacher first gives the student the proposition $\neg(p \rightarrow (q \wedge r))$, then the proposition $p \vee \neg q$, then the proposition $\neg r$, and finally, the proposition $(p \wedge r) \vee (q \rightarrow p)$. For which of the 16 possible sequences of four answers will the student pass the test?
12. Explain why every obligato game has a winning strategy. Exercises 13 and 14 are set on the island of knights and knaves described in Example 7 in Section 1.2.
13. Suppose that you meet three people, Aaron, Bohan, and Crystal. Can you determine what Aaron, Bohan, and Crystal are if Aaron says “All of us are knaves” and Bohan says “Exactly one of us is a knave”?
14. Suppose that you meet three people, Anita, Boris, and Carmen. What are Anita, Boris, and Carmen if Anita says “I am a knave and Boris is a knight” and Boris says “Exactly one of the three of us is a knight”?
15. (Adapted from [Sm78]) Suppose that on an island there are three types of people, knights, knaves, and normals (also known as spies). Knights always tell the truth, knaves always lie, and normals sometimes lie and sometimes tell the truth. Detectives questioned three inhabitants of the island—Amy, Brenda, and Claire—as part of the investigation of a crime. The detectives knew that one of the three committed the crime, but not which one. They also knew that the criminal was a knight, and that the other two were not. Additionally, the detectives recorded these statements: Amy: “I am innocent.” Brenda: “What Amy says is true.” Claire: “Brenda is not a normal.” After analyzing their information, the detectives positively identified the guilty party. Who was it?
16. Show that if S is a proposition, where S is the conditional statement “If S is true, then unicorns live,” then “Unicorns live” is true. Show that it follows that S cannot be a proposition. (This paradox is known as *Löb’s paradox*.)
17. Show that the argument with premises “The tooth fairy is a real person” and “The tooth fairy is not a real person” and conclusion “You can find gold at the end of the rainbow” is a valid argument. Does this show that the conclusion is true?
18. Suppose that the truth value of the proposition p_i is **T** whenever i is an odd positive integer and is **F** whenever i is an even positive integer. Find the truth values of $\bigvee_{i=1}^{100}(p_i \wedge p_{i+1})$ and $\bigwedge_{i=1}^{100}(p_i \vee p_{i+1})$.

- *19. Model 16×16 Sudoku puzzles (with 4×4 blocks) as satisfiability problems.
20. Let $P(x)$ be the statement “Student x knows calculus” and let $Q(y)$ be the statement “Class y contains a student who knows calculus.” Express each of these as quantifications of $P(x)$ and $Q(y)$.
 - a) Some students know calculus.
 - b) Not every student knows calculus.
 - c) Every class has a student in it who knows calculus.
 - d) Every student in every class knows calculus.
 - e) There is at least one class with no students who know calculus.
21. Let $P(m, n)$ be the statement “ m divides n ,” where the domain for both variables consists of all positive integers. (By “ m divides n ” we mean that $n = km$ for some integer k .) Determine the truth values of each of these statements.

a) $P(4, 5)$	b) $P(2, 4)$
c) $\forall m \forall n P(m, n)$	d) $\exists m \forall n P(m, n)$
e) $\exists n \forall m P(m, n)$	f) $\forall n P(1, n)$
22. Find a domain for the quantifiers in $\exists x \exists y (x \neq y \wedge \forall z ((z = x) \vee (z = y)))$ such that this statement is true.
23. Find a domain for the quantifiers in $\exists x \exists y (x \neq y \wedge \forall z ((z = x) \vee (z = y)))$ such that this statement is false.
24. Use existential and universal quantifiers to express the statement “No one has more than three grandmothers” using the propositional function $G(x, y)$, which represents “ x is the grandmother of y .”
25. Use existential and universal quantifiers to express the statement “Everyone has exactly two biological parents” using the propositional function $P(x, y)$, which represents “ x is the biological parent of y .”
26. The quantifier $\exists_n$ denotes “there exists exactly n ,” so that $\exists_n x P(x)$ means there exist exactly n values in the domain such that $P(x)$ is true. Determine the true value of these statements where the domain consists of all real numbers.

a) $\exists_0 x (x^2 = -1)$	b) $\exists_1 x (x = 0)$
c) $\exists_2 x (x^2 = 2)$	d) $\exists_3 x (x = x)$
27. Express each of these statements using existential and universal quantifiers and propositional logic, where $\exists_n$ is defined in Exercise 26.

a) $\exists_0 x P(x)$	b) $\exists_1 x P(x)$
c) $\exists_2 x P(x)$	d) $\exists_3 x P(x)$
28. Let $P(x, y)$ be a propositional function. Show that $\exists x \forall y P(x, y) \rightarrow \forall y \exists x P(x, y)$ is a tautology.
29. Let $P(x)$ and $Q(x)$ be propositional functions. Show that $\exists x (P(x) \rightarrow Q(x))$ and $\forall x P(x) \rightarrow \exists x Q(x)$ always have the same truth value.
30. If $\forall y \exists x P(x, y)$ is true, does it necessarily follow that $\exists x \forall y P(x, y)$ is true?
31. If $\forall x \exists y P(x, y)$ is true, does it necessarily follow that $\exists x \forall y P(x, y)$ is true?

32. Find the negations of these statements.
 - a) If it snows today, then I will go skiing tomorrow.
 - b) Every person in this class understands mathematical induction.
 - c) Some students in this class do not like discrete mathematics.
 - d) In every mathematics class there is some student who falls asleep during lectures.
33. Express this statement using quantifiers: “Every student in this class has taken some course in every department in the school of mathematical sciences.”
34. Express this statement using quantifiers: “There is a building on the campus of some college in the United States in which every room is painted white.”
35. Express the statement “There is exactly one student in this class who has taken exactly one mathematics class at this school” using the uniqueness quantifier. Then express this statement using quantifiers, without using the uniqueness quantifier.
36. Describe a rule of inference that can be used to prove that there are exactly two elements x and y in a domain such that $P(x)$ and $P(y)$ are true. Express this rule of inference as a statement in English.
37. Use rules of inference to show that if the premises $\forall x(P(x) \rightarrow Q(x))$, $\forall x(Q(x) \rightarrow R(x))$, and $\neg R(a)$, where a is in the domain, are true, then the conclusion $\neg P(a)$ is true.
38. Prove that if x^3 is irrational, then x is irrational.
39. Prove or disprove that if x^2 is irrational, then x^3 is irrational.
40. Prove that given a nonnegative integer n , there is a unique nonnegative integer m such that $m^2 \leq n < (m+1)^2$.
41. Prove that there exists an integer m such that $m^2 > 10^{1000}$. Is your proof constructive or nonconstructive?
42. Prove that there is a positive integer that can be written as the sum of squares of positive integers in two different ways. (Use a computer or calculator to speed up your work.)
43. Disprove the statement that every positive integer is the sum of the cubes of eight nonnegative integers.
44. Disprove the statement that every positive integer is the sum of at most two squares and a cube of nonnegative integers.
45. Disprove the statement that every positive integer is the sum of 36 fifth powers of nonnegative integers.
46. Assuming the truth of the theorem that states that $\sqrt{n}$ is irrational whenever n is a positive integer that is not a perfect square, prove that $\sqrt{2} + \sqrt{3}$ is irrational.

Computer Projects

Write programs with the specified input and output.

1. Given the truth values of the propositions p and q , find the truth values of the conjunction, disjunction, exclusive or, conditional statement, and biconditional of these propositions.
2. Given two bit strings of length n , find the bitwise *AND*, bitwise *OR*, and bitwise *XOR* of these strings.
- *3. Give a compound proposition, determine whether it is satisfiable by checking its truth value for all positive assignments of truth values to its propositional variables.
4. Given the truth values of the propositions p and q in fuzzy logic, find the truth value of the disjunction and the conjunction of p and q (see Exercises 50 and 51 of Section 1.1).
- *5. Given positive integers m and n , interactively play the game of Chomp.
- *6. Given a portion of a checkerboard, look for tilings of this checkerboard with various types of polyominoes, including dominoes, the two types of triominoes, and larger polyominoes.

Computations and Explorations

Use a computational program or programs you have written to do these exercises.

1. Look for positive integers that are not the sum of the cubes of nine different positive integers.
2. Look for positive integers greater than 79 that are not the sum of the fourth powers of 18 positive integers.
3. Find as many positive integers as you can that can be written as the sum of cubes of positive integers, in two different ways, sharing this property with 1729.
- *4. Try to find winning strategies for the game of Chomp for different initial configurations of cookies.
5. Construct the 12 different pentominoes, where a pentomino is a polyomino consisting of five squares.
6. Find all the rectangles of 60 squares that can be tiled using every one of the 12 different pentominoes.

Writing Projects

Respond to these with essays using outside sources.

1. Discuss logical paradoxes, including the paradox of Epimenides the Cretan, Jourdain's card paradox, and the barber paradox, and how they are resolved.
2. Describe how fuzzy logic is being applied to practical applications. Consult one or more of the recent books on fuzzy logic written for general audiences.
3. Describe some of the practical problems that can be modeled as satisfiability problems.
4. Explain how satisfiability can be used to model a round robin tournament.
5. Describe some of the techniques that have been devised to help people solve Sudoku puzzles without the use of a computer.
6. Describe the basic rules of *WFF'N PROOF*, *The Game of Modern Logic*, developed by Layman Allen. Give examples of some of the games included in *WFF'N PROOF*.
7. Read some of the writings of Lewis Carroll on symbolic logic. Describe in detail some of the models he used to represent logical arguments and the rules of inference he used in these arguments.
8. Extend the discussion of Prolog given in Section 1.4, explaining in more depth how Prolog employs resolution.
9. Discuss some of the techniques used in computational logic, including Skolem's rule.
10. "Automated theorem proving" is the task of using computers to mechanically prove theorems. Discuss the goals and applications of automated theorem proving and the progress made in developing automated theorem provers.
11. Describe how DNA computing has been used to solve instances of the satisfiability problem.
12. Look up some of the incorrect proofs of famous open questions and open questions that were solved since 1970 and describe the type of error made in each proof.
13. Discuss what is known about winning strategies in the game of Chomp.
14. Describe various aspects of proof strategy discussed by George Pólya in his writings on reasoning, including [Po62], [Po71], and [Po90].
15. Describe a few problems and results about tilings with polyominoes, as described in [Go94] and [Ma91], for example.